

Highlights

Smart Annotation-Aiding Systems: Towards Semantic Sensing in Human-Centric Smart Buildings

First Author, Second Author

- Inquiry-driven annotation framework for residential energy and water services.
- Formalization of occupant inquiries linking sensors, goals, and semantic labels.
- Annotation pipeline combining data collection, interactive labeling, and quality control.
- First-order logic as a shared interface between sensors, reasoners, and LLMs.
- Implementation-oriented evaluation on residential appliance-activity use cases.

Smart Annotation-Aiding Systems: Towards Semantic Sensing in Human-Centric Smart Buildings

First Author^{a,*}, Second Author^b

^aUniv. Grenoble Alpes, CNRS, Grenoble INP, G-SCOP, Grenoble, 38000, France

^bUniv. Grenoble Alpes, CNRS, Grenoble INP, LIG, Grenoble, 38000, France

ARTICLE INFO

Keywords:

smart buildings
occupant behavior
data annotation
residential energy
human–AI interaction
first-order logic
large language models

ABSTRACT

Smart buildings can observe physical consequences of occupant behavior, but they rarely capture the semantic reasons behind those actions. This paper proposes a smart annotation-aiding framework for residential energy and water use, in which occupant-defined inquiries drive the acquisition of sensor evidence and human-provided semantic annotations. Experiments are formalized as inquiry-centered observation processes, and annotations are treated as reusable knowledge rather than disposable training labels. The framework combines inquiry-guided data collection, interactive label assignment, and quality control, and uses first-order logic as a shared representation between sensor-derived facts, symbolic reasoning, and large language models. The approach is illustrated through residential activity use cases and an implementation-oriented evaluation of interactive annotation workflows. The results support annotation-aided sensing as a practical pathway toward more explainable, occupant-centric, and energy-aware building intelligence.

1. Introduction

Most common role of smart buildings is to autonomously optimize energy consumption, comfort, health, and sustainability by continuously observing and reacting to their environment. Modern buildings are increasingly equipped with heterogeneous sensing infrastructures capable of monitoring temperature, humidity, CO₂ concentration, occupancy, power consumption, water usage, and many other physical variables. Despite this abundance of sensing technologies, current smart buildings remain fundamentally limited because they can observe only the physical consequences of occupant behavior, not the reasons that generated them.

A sensor may detect that a window has been opened, a dishwasher has started, or the heating setpoint has increased, but it cannot determine why these actions occurred. The occupant may have opened the window because the air felt stale, because someone was cooking, because of an unpleasant odor, or simply because they preferred fresh air. Likewise, identical electricity consumption profiles may correspond to entirely different human activities, intentions, or comfort expectations. These latent semantic variables are essential for understanding occupant behavior, yet they are inherently inaccessible to physical sensors.

As a consequence, today's smart buildings are primarily reactive systems. They observe measurable events but have only a partial understanding of the human context that produced them. This limitation reduces the accuracy of activity recognition, occupant modeling, demand prediction, personalized control, and explainable decision-making. More fundamentally, it prevents buildings from continuously learning from their occupants.

A truly intelligent building should therefore not only sense its physical environment but also acquire semantic knowledge from its inhabitants. This requires the ability to identify when sensor observations are insufficient, determine which contextual information is missing, interact with occupants at appropriate moments, interpret their responses, and integrate this knowledge into its internal representation of the building state. In other words, a smart building must possess the capability to “listen” to its occupants in the same way that it continuously listens to its sensors.

The proposed smart annotation-aiding systems provide this missing capability. Rather than treating annotations merely as offline labels for machine learning, they establish a permanent communication channel through which occupants can explain intentions, goals, preferences, perceived comfort, satisfaction, or exceptional situations that cannot be inferred automatically. These semantic observations complement physical measurements and transform

*Corresponding author

✉ first.author@example.org (F. Author); second.author@example.org (S. Author)

ORCID(s):

incomplete sensor observations into richer contextual representations of the building state. Such systems are able to overcome the limitations of previous semi-automated data annotation systems in smart-home energy management: revealing the environmental impact of human behavior, reducing the burden of annotation, and enabling more accurate and personalized decision-making.

From this perspective, smart annotation-aiding systems should be considered as a fundamental component of future human-centric buildings. Just as sensors extend a building's ability to perceive the physical world, annotation-aiding systems extend its ability to perceive the human world. Together, these two complementary information sources enable buildings to continuously improve their understanding of occupant behavior, adapt to changing lifestyles, and support more accurate, explainable, and personalized decision-making.

Data annotation is the process of assigning meaningful labels or tags to raw data so that it can be interpreted by machine learning algorithms and used as ground truth for supervised learning (Tan et al., 2024). A data annotation system operationalizes this process through a pipeline that typically includes data collection, label assignment through a defined interface and label schema, and quality control to produce reliable labeled datasets for training and evaluation.

Most annotation systems have been designed with the primary objective of generating ground-truth datasets for supervised machine learning. In this traditional paradigm, annotations represent target labels that are used exclusively during the training phase and are subsequently discarded once the predictive model has been learned. While this approach has proven highly successful across numerous domains, it overlooks the substantial amount of contextual and semantic information that annotations may convey.

An alternative paradigm consists in considering annotations as informative features rather than solely as supervisory labels. In such annotation-aided intelligent systems, annotations become an integral component of the system state, complementing sensor measurements and other automatically acquired variables. Rather than describing only what occurred, annotations may explain why an event occurred, under which circumstances it occurred, or how it was perceived by the human observer. Consequently, annotations provide access to latent contextual variables that are generally inaccessible through physical sensing alone.

The conventional labelling view above is kept only as background: within the proposed framework, an annotation is a *time-related semantic input* that an occupant supplies about a specific sensor context relevant to an inquiry, not a general-purpose feature attached to an arbitrary record. It complements the sensor-based evidence of a case with the meaning the occupant assigns to a particular interval of time, and it is retained as part of that case for later comparison and reasoning rather than discarded once a predictive model has been trained.

This perspective is particularly relevant for cyber-physical systems centred on humans, where many determinants of human behavior remain hidden from conventional sensors. In the context of smart buildings, an occupant annotation such as "I opened the window because the room felt stuffy" contains information about perceived air quality, thermal discomfort, behavioral intention, and adaptive strategies. These semantic concepts cannot be directly inferred from temperature, humidity, or CO₂ measurements, yet they constitute valuable explanatory variables for predicting future occupant actions and energy consumption.

From a machine learning perspective, annotation-aided systems extend the input space by integrating heterogeneous sources of information, including sensor data, human observations, expert knowledge, textual explanations, and confidence indicators. Recent advances in large language models further facilitate this integration by transforming free-text annotations into structured semantic representations or dense vector embeddings that can be combined with conventional numerical features. Such representations enable predictive models to exploit both quantitative measurements and qualitative human knowledge within a unified learning framework.

Data annotation systems are used across many domains, with their design depending on the type of data and task: in *Computer Vision*, for image classification, object detection, and segmentation, especially in healthcare applications (Sager et al., 2021); in *Natural Language Processing (NLP)*, for tasks such as entity recognition, sentiment analysis, and text classification, increasingly with support from Large Language Models (LLMs); and in *Human Activity Recognition (HAR)*, for ambient sensing and IoT-based smart-home data (Demrozi et al., 2023). This study focuses on an extension of HAR: smart-home energy management.

In this domain, sensors can only observe effects, such as power consumption, water usage, door status, or occupancy; they cannot observe the cause behind these effects, namely what the occupant was doing and why. This is because occupant behavior and intent belong to personal experience, not to a physical quantity a sensor can register, no matter how the sensors are configured (del Castillo Cardoso, 2025). Annotation supplies this missing cause: the occupant explains, for example, why a washing machine was started at midnight, or which of several devices sharing the same smart plug (e.g., a TV and a sound system) was actually running.

In addition, deploying annotation in real homes introduces further constraints. Because sensor data are generated continuously, day and night, within the private living space of each household, the annotation system cannot query occupants too frequently, cannot rely on privacy-intrusive sensing, must remain usable by non-expert users, and must adapt to each household's own routines. However, most existing systems only address part of this problem: they either depend entirely on manual occupant input, or automate labeling without adequately supporting context-aware interaction and selective querying.

A number of research directions have proposed partial solutions. Self-experimentation systems address the missing-cause problem by letting occupants pose their own questions, explaining their own routines (del Castillo Cardoso, 2025) and revealing their environmental impact. Interactive and cooperative learning approaches instead target occupant burden, by querying occupants selectively and reconciling labels that conflict with sensor evidence (Silva et al., 2022; Awada et al., 2020). These and other related efforts, along with the gap that remains between them, are examined in detail in the following sections.

This research develops and evaluates a smart annotation-aiding system: an intelligent human–AI collaboration framework that combines sensor observations with human knowledge to acquire, validate, and exploit semantic information that cannot be automatically sensed. Rather than treating annotation as an end goal, the system treats it as one component of a broader knowledge acquisition process that supports learning, prediction, explanation, and decision making. The core objective is to overcome the limitations of previous semi-automated annotation systems in smart-home energy management by suggesting when to annotate, matching concepts shared between human and machine, and proposing labels to reduce annotation burden. Each of the following objectives targets a specific limitation identified above and in the reviewed literature:

1. **Design a goal-based information model:** Mapping data labels to candidate sensor data sources and natural-language question templates, addressing the lack of a systematic way to determine what information is missing from sensor data and how to request it from the occupants.
2. **Build a scenario-aware conversational interface:** Integrating a chatbot capable of dynamically making decisions (to ask, defer, or suppress queries) based on sensor evidence and occupant routines, such as silent modes during late nights or absences, addressing systems that either demand manual input for every event or automate labeling without context-aware, selective interaction.
3. **Define a declarative configuration schema:** Ensuring system flexibility by providing a code-free configuration method, enabling easy scalability to new research domains, addressing the rigidity of existing systems, which are typically built around a fixed set of labels and cannot easily adapt to new households or research domains.
4. **Empirically evaluate the framework:** Conducting real-world experiments to compare the proposed "AnnoBot" with traditional annotation methods on accuracy, user burden, and privacy preservation, validating whether the design choices above actually deliver the reduction in burden and improvement in label quality they are intended to provide.

The study is strictly bounded to residential energy and water consumption analysis, focusing on recognizing occupant behavior and attributing consumption peaks to specific devices or events.

This paper is organized as follows. Section 2 reviews existing annotation systems and related approaches for reducing annotation burden in smart-home and human activity recognition contexts. Section 3 formulates the problem by defining occupant-driven experiments, sensor configurations, temporal horizons, and the annotation system. Section 4 presents the proposed AnnoBot framework and its implementation, including the data collection pipeline, data preprocessing, trigger mechanism, chatbot-based label assignment, and occupant interaction process. Section 5.4 reports the implementation-level results obtained from testing the Telegram-based label assignment module. Finally, Section 6 concludes the paper and discusses future work.

2. Related Works

2.1. Traditional / Manual Annotation Systems

Manual annotation systems rely entirely on the human to decide when and what to label. A representative example is Label Studio, one of the most widely used open-source labeling platforms, adopted by leading technology organizations such as Meta, NVIDIA, and Intel. Its mechanism follows the manual paradigm: every instance imported into the system is treated as requiring a label, and it is left to the annotator to decide which task to work on and when, with no automatic mechanism determining whether a given instance is actually worth labeling (HumanSignal, Inc., n.d.). This design is

well suited to enterprise settings in the CV or NLP domains, where formally trained annotators are assigned to process a fixed, pre-collected dataset, but it makes such systems difficult to apply directly to the HAR domain, where data are continuous and annotation takes place within the residents' own homes, by individuals who lack professional labeling experience.

In the residential HAR setting, the interface is simplified for non-expert users, and occupants are prompted to label events as they occur in the home. A simple labeling system for elderly individuals used physical push-buttons installed at locations where the corresponding activities typically take place, so that pressing a button directly assigns a label to the ongoing activity (Matsui et al., 2020). This approach is easy to use and well suited to users unfamiliar with technology, but it is limited to predefined activities (bathing, cooking, eating, going out, sleeping, etc.), and label accuracy depends entirely on whether the occupant remembers to press the button at the correct start and end times of the activity.

Another study extended this scope further with a recording interface that allowed up to 27 different activities to be recorded, producing 60 to 100 labels per day over a 28-day trial (Alemdar et al., 2013). Although considerably more flexible than fixed push-buttons, this approach still requires the occupant to actively act on every single event, with no mechanism to filter or prioritize the events that genuinely warrant annotation.

Taken together, whether general-purpose or simplified for residential use, these systems retain the core mechanism and limitation of the manual category: the decision of when and what to annotate depends entirely on the human, with no support from the system itself. In the context of sensor data generated continuously from the residential environment, this can render the annotation workload excessive in the absence of an intelligent filtering and prioritization mechanism; this is precisely the motivation for the semi-automated systems presented in the following section.

2.2. Semi-Automated Systems: Application of Interactive Learning in Annotation System

Interactive learning has been studied as a way to reduce annotation effort by allowing models to selectively use human feedback when labels are uncertain, ambiguous, or costly to obtain. In annotation systems, this paradigm shifts labeling from exhaustive manual annotation toward targeted human intervention.

Prior HAR surveys provide useful background for this direction. Bouchabou et al. (2021a) highlight challenges related to sparse sensing, context dependence, and deployment variability, while Hussain et al. (2019) provide a broader taxonomy of sensing paradigms and recognition challenges. Although these works are not centered on annotation systems, they motivate the need for efficient, adaptive, and human-in-the-loop labeling methods.

2.3. Methods for Reducing Annotation Burden

Several studies have explored ways to reduce annotation effort in smart-home and activity-related systems. Study (Silva et al., 2022) are the closest to this project's direction, as they propose a hybrid interactive and cooperative learning framework that combines selective querying, user-defined labels, and confusion repair. Rather than treating annotation as a one-time labeling task, their system frames it as an adaptive dialogue where the model asks for feedback when needed and revisits inconsistent past decisions.

Even with user participation, labeling errors, subjectivity, or inconsistencies can still occur. A Cooperative Learning mechanism addresses this by detecting contradictions between a label just provided by the user and the actual sensor measurements, or against previously recorded labels; rather than discarding the data automatically, the system sends a notification suggesting a correction for the user to confirm, amend, or reject (Awada et al., 2020). This two-way mechanism progressively aligns the system's records with the user's actual real-world experience.

Other works reduce annotation burden by changing how supervision is collected. DISCOVER (Karpekov et al., 2025) clusters recurring sub-activities from unlabeled event streams and asks humans to label representative centroids instead of dense raw sequences. Accoustate (Chatterjee et al., 2021), although not directly designed for ambient smart homes, also reflects the broader concern of reducing manual labeling through modality-assisted auto-annotation. Together, these studies show that annotation-efficiency methods remain fragmented across interactive querying, structure-based label, compression, and assisted auto-annotation.

Another related direction focuses on semantic adaptation and cross-home generalization. Bouchabou et al. (2021b) use language-model-based semantic enrichment to represent sensor events beyond raw identifiers. TDOST (Thukral et al., 2024) further converts sensor triggers into textual descriptions, allowing models to transfer across homes with different layouts and sensor vocabularies. DomusFM (Fiori et al., 2026) extends this idea through smart-home foundation-model pretraining across multiple datasets. These works improve deployability under limited labels, but they still do not fully solve the problem of aligning transferred representations with user-specific and evolving labels.

Explainability is also relevant to annotation systems because users are more likely to correct or refine labels when model outputs are understandable. Das et al. (2021) show that explanation design affects user trust in smart-home recognition systems. GNN-XAR (Fiori et al., 2025) incorporates interpretability into a graph-based recognition model, while Fiori et al. (2024) explore the use of large language models to evaluate explanation quality. These studies improve transparency, but they do not directly address when the system should ask for feedback or how conflicting labels should be repaired.

Finally, contextual reasoning methods address ambiguity when local sensor evidence is insufficient. TRACE (Shi et al., 2026) is especially relevant because it uses longer-horizon context and user-specific priors to produce more coherent predictions under sparse sensing and domain variation. Compared with Silva et al. (2022), which emphasizes human involvement during annotation, TRACE focuses on reducing ambiguity within the inference process itself. These directions are complementary: effective annotation systems likely require both contextual reasoning to reduce uncertainty and interactive feedback to resolve the uncertainty that remains.

2.4. Research Gaps

Across the reviewed literature, the main gap is no longer a lack of awareness that smart-home HAR and annotation are difficult. Existing review papers have already established the core challenges, including sparse sensing, context dependence, multi-user complexity, concept drift, and poor cross-home transferability (Bouchabou et al., 2021a; Hussain et al., 2019). Meanwhile, recent method-oriented studies provide partial solutions from different perspectives, such as reducing annotation effort, improving semantic transfer, enhancing explainability, and strengthening contextual reasoning. However, these directions remain largely fragmented and do not yet form an integrated annotation framework for user-specific, evolving smart-home environments.

Table 1: Comparison of related studies and remaining gaps for smart-home annotation systems.

Research Direction		Representative Studies	Main Contribution	Remaining Gap
Interactive cooperative annotation	and	Silva et al. (2022); Awada et al. (2020)	Reduce labeling effort through selective querying, user-defined labels, interactive feedback, and cooperative correction of inconsistent labels.	Interaction is mainly used for label acquisition or correction.
Structure-based label reduction		DISCOVER (Karpekov et al., 2025)	Clusters recurring sub-activities from unlabeled sensor streams and asks humans to label representative centroids instead of dense raw sequences.	Reduces annotation volume, but human feedback is still separated from continuous, context-aware interaction during daily deployment.
Assisted annotation	auto-	Accoustate (Chatterjee et al., 2021)	Uses modality-assisted auto-annotation to reduce manual labeling requirements.	Demonstrates the broader value of assisted supervision, but is not designed specifically for ambient smart-home annotation with privacy-preserving environmental and power sensors.
Semantic adaptation and cross-home transfer		Language-model bootstrapping (Bouchabou et al., 2021b); TDOST (Thukral et al., 2024); DomusFM (Fiori et al., 2026)	Enrich sensor events with semantic descriptions or foundation-model representations to improve transfer across homes, layouts, and sensor vocabularies.	Improves deployability under limited labels, but does not fully align transferred representations with user-specific routines, evolving labels, and interactive correction.
Explainability and trust		Das et al. (2021); GNN-XAR (Fiori et al., 2025); Fiori et al. (2024)	Improve transparency of smart-home recognition systems and study how explanations affect user trust and understanding.	Explanations help users understand model outputs, but these works do not directly determine when feedback should be requested or how conflicting annotations should be repaired.
Contextual reasoning under ambiguity		TRACE (Shi et al., 2026)	Uses longer-horizon context and user-specific priors to improve prediction consistency under sparse sensing and domain variation.	Reduces inference ambiguity, but does not focus on interactive annotation or on incorporating user feedback to revise uncertain or incorrect labels.

As shown in Table 1, what is still missing is a unified annotation-centered framework that integrates these strengths into a single human-in-the-loop process. In particular, existing studies still lack:

- **Integrated annotation workflow:** most studies optimize only one aspect of the problem, such as reducing the number of labels, improving transferability, enhancing explanations, or strengthening contextual reasoning, rather than combining these capabilities in one annotation pipeline.
- **Cooperative ambiguity repair:** although user feedback is used in some systems, fewer works support a two-way mechanism for detecting, revisiting, and repairing inconsistent or ambiguous labels over time.
- **Privacy-preserving interaction:** many approaches do not explicitly focus on annotation using non-intrusive ambient environmental and power sensors, which are important for preserving privacy in residential environments.

- **User- and home-specific semantic adaptation:** transferred or generalized models are not fully aligned with the evolving routines, label meanings, and contextual preferences of individual occupants and homes.

These limitations motivate the development of our framework, *AnnoBot*: a flexible LLM-based chatbot integrated with dialogue policy mechanisms for adaptive, privacy-preserving, and user-centered annotation, as discussed in the following sections.

3. Problem Formulation

Following the assumption by del Castillo Cardoso (2025), occupants are not only passive recipients of automated smart-home services, but can also act as intentional learners who actively question, observe, and reflect on their own residential behaviors. In everyday life, occupants may naturally wonder how much energy is consumed by household activities, how specific appliances or services operate, and how changes in these services may affect both energy consumption and indoor environmental quality. For example, an occupant may ask whether a certain washing-machine mode is more energy-efficient, whether opening windows at specific times improves comfort, or whether a particular routine increases electricity usage.

3.1. From self-experimentation to occupant-driven inquiry in smart homes

The idea that individuals can generate scientific knowledge by experimenting on themselves has a long history. Self-experimentation has been widely practiced in medicine, physiology, and psychology, where researchers voluntarily become the subjects of their own investigations. Although this practice dates back several centuries, it was first comprehensively analyzed by (?), who described self-experimentation as an important yet underappreciated scientific tradition. In this context, the objective of self-experimentation is not merely to collect observations but to test hypotheses under controlled conditions while reducing experimental uncertainty. More recently, self-experimentation has re-emerged through the Quantified Self movement (?) and personal science (?), where individuals use digital technologies to investigate questions about their own health, behavior, or lifestyle. These approaches emphasize that scientific inquiry can be conducted by non-experts using personal data collected in everyday life.

The proposed framework shares the principle that the individual is both the investigator and the subject of the investigation, but extends it beyond personal health toward domestic energy services. Rather than studying physiological responses, occupants investigate how their own activities influence comfort, building services, and resource consumption within their everyday living environment. The home therefore becomes an instrumented environment supporting systematic observation and experimentation.

This perspective is closely related to Dewey's theory of inquiry (?), in which knowledge emerges from the investigation of an indeterminate situation through observation, experimentation, and reflection. Inquiry is initiated by a meaningful question arising from practical experience and progresses through the deliberate collection of evidence until a satisfactory explanation is obtained. Rather than prescribing optimal behaviors, inquiry supports understanding by enabling individuals to relate their actions to their observed consequences. This pragmatic view provides an appropriate conceptual foundation for occupant-centred smart-home systems because the objective is not to optimize behavior automatically but to support occupants in understanding their own energy practices.

Current smart-home research adopts a substantially different perspective. Most energy management systems rely on sensor data, predictive models, and optimization algorithms to automatically control building services or recommend energy-efficient actions. Occupants are generally considered as passive data providers or recipients of recommendations, while annotations are primarily used to construct labelled datasets for activity recognition, occupancy detection, or supervised learning. Consequently, data acquisition is driven by the requirements of machine learning models rather than by the information needs of occupants.

The proposed framework departs from this paradigm by organizing data acquisition around an *Experiment*: a time-bounded listening period, set up by an occupant, that groups one or more concrete *Inquiries*. Each inquiry originates from a question that the occupant wishes to answer and specifies the observations, contextual information, and semantic annotations required to answer that question within the experiment's listening period. Sensor measurements therefore no longer constitute an end in themselves but become evidence supporting a structured investigation. Similarly, annotations are not merely labels for training predictive models but semantic explanations that contextualize observations and enable their interpretation. This shift transforms the smart annotation-aiding system from a passive data collection framework into an interactive knowledge acquisition process, where occupants progressively build an understanding of what constitutes an appropriate, acceptable, and sufficient level of service within their own homes.

This perspective naturally introduces the notion of an inquiry: an occupant-driven investigation, configured inside an experiment, centred on a question they wish to answer. Typical questions could be how much lowering a thermostat setpoint affects perceived comfort and energy consumption, or whether an alternative dishwasher programme remains satisfactory while reducing energy consumption, or whether shifting appliance operation to another period influences household routines. A single experiment can group several such inquiries under one listening period; for instance, the same two-week meeting-room experiment can simultaneously investigate several distinct questions about occupancy, energy sources, comfort trade-offs, or anomalies (Section 5.2).

An inquiry provides the semantic context that gives meaning to sensor observations. Instead of continuously acquiring all available data, the system collects only the information specified by the occupant to answer their question, i.e., which sensors should be monitored, which quantities should be compared or aggregated to assess the outcome, how a relevant service episode should be detected and bounded within the experiment's listening period, and which semantic dimensions require annotation. For instance, an inquiry about the washing machine detects one washing-machine cycle as its service episode, while an inquiry about the HVAC system relies on a listening period spanning one day, from 3am to 3am the next day, inherited from its experiment. It therefore guides the complete information acquisition process, from sensor selection and cycle detection to annotation and data interpretation.

Each inquiry is realized through its experiment's listening period: sensor observations constitute empirical evidence, annotations provide semantic explanations when required, and the resulting sensor-based or annotated situations become cases that can later be compared with similar inquiries. In this way, an experiment transforms isolated measurements into interpretable knowledge that supports learning, explanation, and informed decision-making across all of its inquiries.

This organization differs fundamentally from traditional data annotation systems. Conventional annotation frameworks are centred on datasets whose primary objective is to produce labelled examples for supervised learning. In contrast, the proposed framework is organized around occupant-defined inquiries, each configured inside an experiment, designed to answer meaningful questions about domestic energy services. Annotation is therefore not an end in itself, but one component of a broader inquiry process through which occupants progressively discover what constitutes an appropriate, acceptable, and sufficient level of service within their own homes.

An experiment groups the inquiries an occupant configures during one overall listening period:

$$E = \langle I, T \rangle, \quad I = \{i_1, \dots, i_n\}, \quad (1)$$

where T is the experiment-level listening period (Section 3.1.4) and each $i_j \in I$ is one inquiry (question) configured within the experiment; I denotes the set of inquiries, not a set of indicators.

Each inquiry conducted by an occupant to investigate an empirical energy-related question in a real residential environment is formally defined as the tuple

$$i = \langle Q, S, I, D, A, \Gamma \rangle, \quad (2)$$

where Q denotes the inquiry question expressed in natural language, S represents the selected sensors configuration, I denotes the indicators used to compare or assess the outcome, D specifies how a relevant service episode (cycle or case) is detected and temporally bounded within the experiment's listening period, A denotes the semantic annotation scope required beyond what S can provide, and Γ denotes the inquiry goal, that is, the underlying hypothesis or decision problem that the inquiry aims to address. The components Q , S , D and I are formally described in Sections 3.1.1, 3.1.2, and 3.1.3, respectively; the experiment-level listening period T is described in Section 3.1.4.

Distinguishing Q from Γ is important: Q is the natural-language formulation posed by the occupant, whereas Γ captures the operational objective that later drives annotation triggers, chatbot interaction, and case retrieval. Table 2 illustrates this distinction with representative examples.

Table 2: Examples of experiment questions Q and corresponding inquiry goals Γ .

Experiment question Q	Inquiry goal Γ
Which washing mode is more economical?	Compare washing modes
Can I reduce the setpoint by 1°C?	Evaluate comfort-energy trade-off
Why was yesterday's electricity consumption unusually high?	Explain an anomaly

The purpose of the experiment is not to control the occupant's behavior automatically, but to support the occupant in collecting relevant evidence, observing the consequences of everyday actions, and gradually understanding the relationship between residential context, appliance usage, energy consumption, and comfort-related outcomes. This assumption is consistent with the notion of intentional learners, where occupants actively participate in understanding their own living environment rather than merely following system-generated recommendations.

Data are collected into cases. A case is a collection of sensor observations, contextual information, and semantic annotations that are relevant to the inquiry goal. A case is built up as

sensor trace → detected cycle / service episode → case evidence → (optional accepted annotations) → annotated case,

where the annotation step is optional: when the inquiry question can be answered from sensor evidence alone, the detected episode already constitutes a valid, sensor-based case, and it becomes an annotated case only once accepted semantic annotations are attached to it. Cases are used to compare and analyze the results of different experiments. They can be displayed to the occupant to help them understand the relationship between their actions and the outcomes of their experiments (?), or used to develop a case-based reasoning system to answer questions about future actions (?), etc. . .

3.1.1. Experiment Question

The inquiry question Q defines the main empirical inquiry that the occupant aims to investigate. It guides which information is relevant to collect and determines the level of semantic detail required to characterize residential facts. In this framework, Q is not only a textual question, but also a mechanism for delimiting the annotation scope, since occupants may choose which semantic dimensions are necessary for their experiment, such as the 5W1H dimensions: Who, What, Where, When, Why, and How.

The definition of Q also determines how annotations are processed. Depending on the required level of contextual information, the framework distinguishes between two recognition strategies:

1. **Annotation-free recognition:** When the inquiry question can be answered using sensor data alone, explicit occupant annotations are not required. In this case, raw sensor streams are processed through *signature extractors*, which transform sensor measurements into interpretable representations of facts, such as detecting whether an appliance is ON or OFF.
2. **Annotation-based recognition:** When the inquiry question requires contextual or semantic information that cannot be inferred directly from sensors, occupant annotations become necessary. For example, a sensor may detect that a washing machine is operating, but it cannot determine the selected washing program or the occupant's intention. In such cases, the system requests semantic annotations to enrich the sensor-based facts.

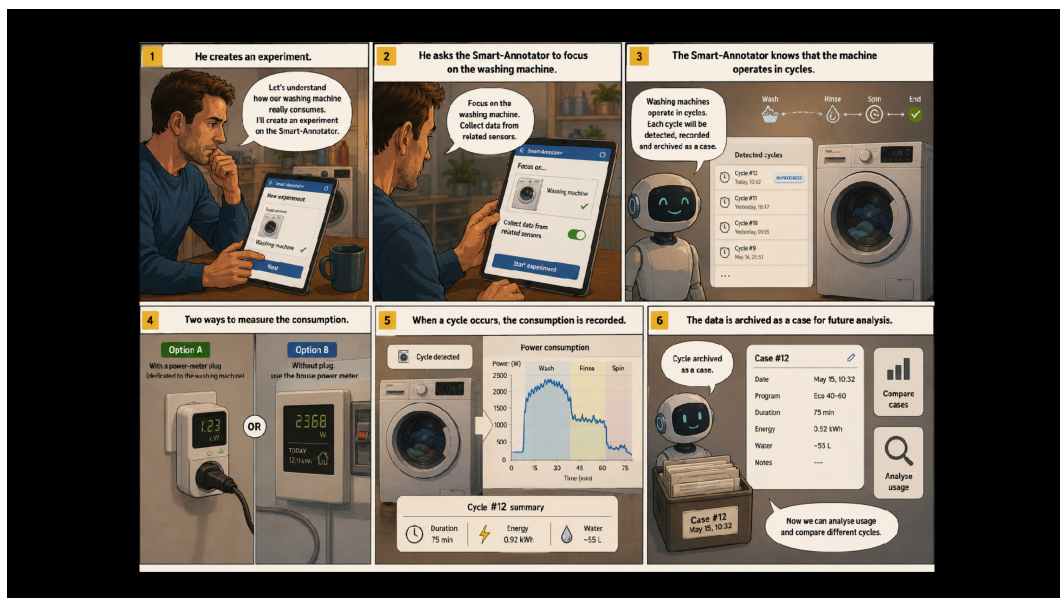


Figure 1: Illustrative use case of defining a washing-machine experiment and archiving detected washing cycles as cases for later analysis.

Figure 1 illustrates a washing-machine use case in which an occupant creates an inquiry, selects the washing machine as the target appliance, and allows the system to collect data from related sensors. The system then detects washing cycles from appliance operation patterns, records consumption-related attributes, and archives each detected cycle as a case for future comparison and analysis.

For example, consider the same residential activity: using the washing machine. If the occupant asks:

When does the washing machine operate, and how much energy does each cycle consume?

The required annotation scope is relatively narrow. The system mainly needs to detect washing-machine operation from power-consumption signatures, segment each cycle, and archive factual attributes such as start time, end time, duration, and energy consumption. In this case, the relevant dimensions are mostly *What*, *When*, and possibly *How*, where *How* refers to measurable operational patterns such as power profile or cycle duration. Since these facts can be derived from sensor data, the system may adopt an annotation-free recognition strategy.

In contrast, if the occupant asks:

For regular laundry and bedding laundry, which washing mode is more energy-efficient?

The annotation scope becomes broader. The system may still detect the same washing-machine cycles from sensor data, but answering this question now requires additional semantic information that sensors alone cannot reliably provide. For instance, the system needs to know whether each cycle was used for regular clothes or bedding items such as blankets, pillows, or mattress covers. It may also need to record the selected washing mode, load type, and possibly load size in order to compare energy consumption across different usage contexts. In this case, the relevant dimensions expand beyond *What* and *When* to include more detailed forms of *How*, such as the washing mode, and *Why*, such as the purpose of the cycle. Therefore, occupant annotation becomes necessary.

Therefore, Q plays a central role in both defining the purpose of the experiments and selecting the appropriate annotation strategy, while Γ keeps the downstream interaction and retrieval mechanisms aligned with the underlying inquiry goal.

3.1.2. Sensors Configuration

The sensors configuration S defines the set of sensors selected to support a given inquiry. Rather than considering all sensors deployed in the residence, each experiment focuses only on the sensors that are relevant to the occupant's question.

The definition of S determines which observations are collected and processed for the experiment. This allows different empirical questions to be studied independently, each with its own appropriate set of sensors and corresponding characterization. For example, an experiment about the energy impact of washing machine configurations may focus on laundry-related sensors, while an experiment about cooking-related energy use may rely on kitchen appliances and environmental sensors.

Figure 1 illustrates this configuration through the washing-machine use case. Once the occupant defines an inquiry about washing-machine usage, the system does not need to collect data from all household sensors. Instead, it focuses on laundry-related sensors that can provide evidence about the operation of the washing machine. Depending on the available installation, this may include a dedicated power-meter plug attached to the washing machine, or a household-level power meter from which washing-machine signatures can be extracted.

Thus, S constrains the data acquisition and analysis process by selecting only the sensor information required to answer the inquiry question.

3.1.3. Cycle or Case Detection and Indicators

D specifies how a relevant service episode, or cycle, is detected and temporally bounded inside the experiment's listening period. Detection returns the episode boundaries t_{start} and t_{end} from the observed sensor traces; for a washing machine or dishwasher, a power profile lets a smart plug identify when the appliance started and stopped operating.

I is distinct from D : while D identifies and bounds the episode, I defines the quantities used to compare or assess it. For a power signal $P(t)$ observed over the interval returned by D , the cycle energy indicator is

$$E_{\text{cycle}} = \frac{1}{3600} \int_{t_{\text{start}}}^{t_{\text{end}}} P(t) dt \quad [\text{Wh}], \quad (3)$$

with P expressed in watts and t in seconds. Detection therefore supplies the integration bounds, while the indicator computes the comparison value; a single inquiry may define several indicators over the same detected episode, for example the peak power and the total energy consumed.

3.1.4. Temporal Horizon

The experiment start and stop conditions T^- and T^+ define the overall period during which the experiment is observed and analyzed. This period, referred to as the *listening period*, is configured once for the experiment and shared by all of the inquiries it contains: it allows occupants to specify when the system should collect data relevant to any of those inquiries. It can be defined in terms of start and end dates, selected days of the week, and specific intra-day time intervals, consumption thresholds, triggering annotations, etc. . . .

The experiment-level listening period must not be confused with a detected cycle or service episode: the listening period can span, for example, an entire two-week meeting-room study, whereas each individual meeting or appliance cycle within it has its own, much shorter, $[t_{\text{start}}, t_{\text{end}}]$ returned by D (Section 3.1.3).

3.2. Annotation System

An inquiry defines *what* should be investigated; the annotation system defines *how* the corresponding evidence is acquired, enriched, and validated. In the proposed framework, an inquiry is realized through its experiment's listening period, during which occupants pose questions about domestic energy services, observe the consequences of their routines, and progressively relate actions to comfort and consumption outcomes (del Castillo Cardoso, 2025). The annotation system supports this process by turning sensor streams into interpretable cases that combine measurements with semantic explanations, following the sensor trace $\rightarrow$ detected episode $\rightarrow$ case evidence $\rightarrow$ annotated case sequence introduced in Section 3.1.

As introduced in Section 3.1.1, the required annotation strategy depends on Q and Γ , giving rise to annotation-free recognition when Q can be answered from sensor-derived facts alone, and annotation-based recognition when it also requires contextual or intentional information. This paper focuses on the second regime: sensors still provide the empirical backbone of each experiment, but occupant feedback supplies the missing semantic dimensions demanded by Γ .

Operationally, the annotation system is a pipeline with three components: data collection, label assignment, and quality control. Data collection gathers the sensing evidence selected by S during the listening window $[T^-, T^+]$; label assignment attaches the semantic dimensions required by Q ; and quality control decides whether the resulting annotations are reliable enough to enter the case base. These components are formalized below and developed methodologically in Section 4.

3.2.1. Data Collection

Data collection acquires observations from the sensor set S associated with the inquiry. Because residential sensing setups differ, sensors may report events under different mechanisms (Table 3). Periodic sensors provide a regular view of continuous variables, whereas trigger-based sensors report state changes that often mark activity boundaries. Both are useful for inquiry: the former characterizes context, the latter highlights candidate moments for annotation.

Table 3: Comparison of sensor event reporting mechanisms.

Mechanism	Description	Examples
Periodic event	Observations are sent at an approximately fixed sampling interval.	Temperature, humidity, CO ₂ , or power readings every few minutes.
Trigger-based event	Observations are sent only when a condition or state change occurs.	Motion detection, door opening, appliance ON/OFF transitions, or sudden power changes.

Let $S = \{s_1, s_2, \dots, s_n\}$ denote the selected sensors. The event history collected over the listening window is

$$D_T = \{e_j \mid e_j = \langle s_i, \tau_j, v_j, m_j \rangle, s_i \in S, \tau_j \in [T^-, T^+]\}, \quad (4)$$

where s_i is the source sensor, τ_j the timestamp, v_j the observed value, and m_j the measurement type (e.g., power, temperature, motion, or contact state).

Persistence is part of the definition of data collection, not an implementation detail. Events are stored so that later stages can reconstruct the same experiment, aggregate readings into timeslots, decide whether to ask the occupant, and revise labels if needed. In other words, D_T is the empirical substrate from which cases are built.

3.2.2. Label Assignment

Label assignment associates semantic information with the timeslot representation derived from D_T . After aggregation, let X_T denote the timeslot-sensor matrix whose k -th row $\mathbf{x}_k$ describes the sensing context of timeslot W_k . A label for that timeslot is a structured semantic description

$$y_k = \langle y_k^{\text{who}}, y_k^{\text{what}}, y_k^{\text{where}}, y_k^{\text{when}}, y_k^{\text{why}}, y_k^{\text{how}} \rangle, \quad (5)$$

corresponding to the 5W1H dimensions. Not every inquiry needs every dimension. The question Q delimits the annotation scope, while Γ indicates which dimensions are decisive for the decision or comparison at stake. For example, comparing washing modes may require *what* and *how*; explaining an anomalous consumption peak may additionally require *why*.

The result of this stage is the set of annotated timeslots

$$\mathcal{A} = \{(\mathbf{x}_k, y_k) \mid \mathbf{x}_k \in X_T, y_k \text{ is available}\}. \quad (6)$$

Each pair $(\mathbf{x}_k, y_k)$ links a sensor-based context to an occupant-provided explanation and is the annotation-based case representation once y_k has been accepted through quality control: $\mathbf{x}_k$ says what the sensors observed, and y_k says how the occupant interprets that observation with respect to the inquiry. This pair representation applies only when the inquiry requires annotation-based recognition; under annotation-free recognition, $\mathbf{x}_k$ alone already constitutes a valid sensor-based case, so not every case requires a y_k .

3.2.3. Quality Control

Because annotations are provided by non-expert occupants in real homes, they may be missing, incomplete, inconsistent with prior cases, or uncertain. Quality control therefore evaluates whether an annotated sample is ready to support the inquiry before it enters the validated case base.

For $(\mathbf{x}_k, y_k) \in \mathcal{A}$, the system computes

$$q_k = \psi(\mathbf{x}_k, y_k), \quad (7)$$

where $\psi(\cdot)$ inspects properties such as completeness of the dimensions required by Q , consistency with similar past situations, and the need for further clarification. The validated set is

$$\mathcal{A}^* = \{(\mathbf{x}_k, y_k) \in \mathcal{A} \mid \psi(\mathbf{x}_k, y_k) = \text{valid}\}. \quad (8)$$

Samples that fail these checks are not simply discarded. They may be marked as incomplete, uncertain, or conflicting and returned to the interaction loop for confirmation or refinement. Quality control thus protects both downstream learning and the integrity of the inquiry itself: only annotations that are sufficiently reliable become part of the evidence used to answer Q under goal Γ .

Figure 2 summarizes how the components introduced above chain together end to end; Sections 4.2–4.4 then detail how first-order logic, the LLM, and the interactive/cooperative learning mechanisms implement each step.

4. Proposed Framework

In this study, with the objective of supporting occupant-centered annotation for residential energy-related experiments, we propose an annotation framework composed of three main modules: *Data Collection*, *Label Assignment*, and *Quality Control*. The overall goal is to transform raw sensor observations into timeslot-level representations and enrich them with semantic labels through interactive occupant feedback.

4.1. Data Collection

The first module, *Data Collection*, acquires the empirical evidence required by an inquiry. Given an inquiry $\langle Q, S, T^-, T^+, \Gamma \rangle$, only the sensors in S are monitored during the observation window delimited by T^- and T^+ ; data

acquisition is therefore inquiry-driven rather than exhaustive, collecting only the signals needed to support Q and Γ instead of the whole household. As shown in Fig. 3, physical devices are not accessed directly: a retrieval layer exposes them through a query interface, normalizes their readings, and persists them so that later stages can reconstruct the selected sensing context. The implementation of this layer is described in Sections 4.1.1 and 4.1.2.

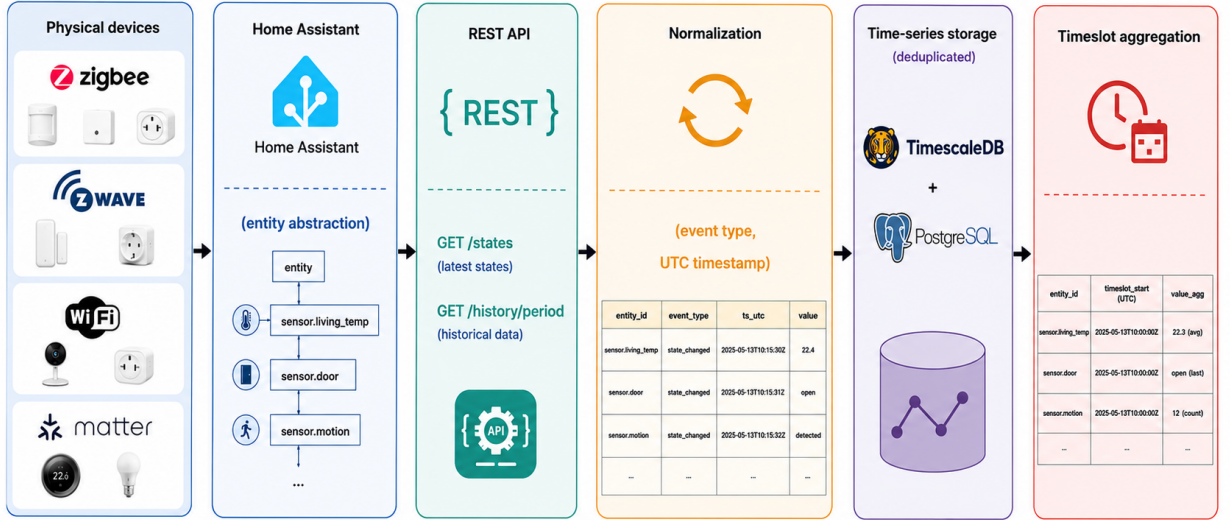


Figure 3: Data Collection pipeline, from physical sensors to timeslot-level representations.

4.1.1. Sensor Registration and Data Retrieval

Physical devices in the residence, such as smart plugs, motion sensors, contact sensors, or environmental sensors, may connect using heterogeneous wireless protocols, including *Zigbee*, *Z-Wave*, *Wi-Fi*, or *Matter*. Rather than implementing a separate retrieval pipeline for each protocol, the deployed system connects to physical devices through Home Assistant (Home Assistant, n.d.), which abstracts these protocols and exposes device information and sensor readings through a unified REST API. The REST API is preferred over the WebSocket interface that Home Assistant also offers because it exposes a broader set of endpoints and is simpler to integrate into the backend, which is implemented with FastAPI.

Since the REST API does not push updates in real time, the backend instead polls Home Assistant on the schedule shown in Table 4; occupants can also trigger an on-demand refresh from the dashboard. To avoid re-fetching history on every cycle, each sensor resumes synchronization from a checkpoint of its last retrieved reading, and a newly added sensor is backfilled over the past week.

Table 4: Synchronization schedule between the backend and Home Assistant.

Synchronization target	Frequency
Device registry	Every 30 minutes
Sensor readings	Every 1 minute
On-demand refresh	Triggered manually by the occupant from the dashboard

4.1.2. Data Normalization and Storage

Readings retrieved from Home Assistant differ in naming conventions, units, attributes, and timestamp formats, and are therefore converted into a common event representation before being stored, as summarized in Table 5. Attributes that cannot be mapped to a known sensor or event type are discarded rather than stored.

Table 5: Normalization applied when converting Home Assistant readings into stored sensor events.

Field	Home Assistant representation	Normalized representation
Sensor / event type	device_class or unit (power, °C, L/min, ...)	event_type (power_w, temp_c, ...) and sensor_type (power_meter, motion, ...)
Value	state as a string (e.g. "235.5")	value cast to float (e.g. 235.5)
Timestamp	last_changed, local time	event_time, UTC
Identifier	entity_id (e.g. sensor.z1_power...)	source; entities of the same device are merged
Filtering	Unmapped device_class / unit	Discarded

After normalization, device metadata is stored in a regular PostgreSQL table, while sensor readings are stored in a TimescaleDB hypertable keyed by the composite of `event_time` and `id`, as detailed in Table 6. This allows readings to be retrieved through time-oriented operators such as `time_bucket()`, `first()`, `last()`, or `sum()`, while TimescaleDB’s built-in compression and retention policies keep long-term storage costs manageable as the event history grows.

Table 6: Schema of the sensor-event hypertable.

Column	Type	Description
<code>event_time / id</code>	TIMESTAMPTZ / VARCHAR(36)	Composite primary key: reading timestamp and record UUID.
<code>source</code>	VARCHAR(128)	Originating Home Assistant <code>entity_id</code> .
<code>event_type</code>	VARCHAR(64)	Normalized event type.
<code>value</code>	DOUBLE PRECISION	Normalized reading value.
<code>metadata</code>	JSONB	Optional attributes, GIN-indexed.
<code>created_at / updated_at</code>	TIMESTAMPTZ	Bookkeeping timestamps, default to insertion time.

The collected stream nevertheless remains a low-level event history. Before it can support annotation, dialogue with occupants, or symbolic reasoning, it must be transformed into a representation that is both computationally usable and semantically interpretable; this transformation is addressed next.

4.2. Data Preprocessing

Residential sensors typically produce irregular, heterogeneous streams: some report periodically, others only on state changes, and their units and sampling rates differ. Preprocessing therefore converts the normalized event history stored in the time-series database into a regular timeslot representation and then into discrete semantic values that can be shared with occupants and with higher-level reasoning components.

4.2.1. Discretization of sensor data

Let t_k denote the k -th timeslot. For each sensor $s_i \in S$, the system builds a time series $s_i(t_k)$ over $t_k \in [T^-, T^+]$. When useful for the inquiry, aggregate indicators are also derived, such as the energy consumed by an appliance, a room, or the dwelling over the experiment.

Continuous measurements are then mapped to discrete domains $d_i \in D_i$ by an adjustable function ϕ_i^θ . Discretization is not introduced only for compression. In the proposed framework it serves three purposes:

- **Human interpretability:** occupants reason more naturally in terms of *comfortable* or *high consumption* than in terms of raw Celsius or watt values;
- **Language grounding:** discrete levels provide a stable vocabulary for chatbot interaction and for LLM prompts;
- **Symbolic lifting:** discrete values can be turned into first-order predicates without committing prematurely to brittle numerical thresholds inside logical rules.

Table 7 gives representative domains.

Table 7: Examples of discrete domains after sensor-data discretization.

Sensor / indicator	Example discrete values
Power at t_k	none, low, medium, high, very_high (or one total-consumption level after summing over the experiment)
Indoor temperature at t_k	very_cold, cold, slightly_cold, comfortable, slightly_hot, hot, very_hot (or one averaged level over the experiment)
Motion	no_motion, motion
Door / window	open, closed
Light	on, off
Presence / sound	detected, not_detected

Because perception is household-specific, the thresholds θ are adjustable. An occupant who contests a reported level—for example, rejecting a temperature labelled as *comfortable*—provides feedback that can recalibrate ϕ_i^θ . Discretization thus participates in the inquiry loop: it translates physical measurements into shared semantic categories that both the occupant and the system can discuss, revise, and reuse.

4.2.2. First-order logic representation

Discretized observations remain fragmented unless they are organized as explicit knowledge. The framework therefore encodes each timeslot with first-order logic (FOL) formulas. FOL is chosen because it can express objects, relations, and rules in a form that is machine-checkable, explainable, and sufficiently textual for LLMs to consume.

A formula is built from symbols that denote:

- objects (people, rooms, appliances),
- events and activities (cooking, sleeping, opening a window),
- relationships (presence in a room, use of an appliance),
- contextual or causal knowledge relevant to the inquiry.

Typical predicates include

- DoorOpen(*door*, *time*),
- PowerConsumption(*device*, *value*, *time*),
- Cooking(*person*, *time*),
- Using(*person*, *device*, *time*).

Functions such as RoomOf(*device*) or Duration(*activity*) may enrich these statements.

Sensor-derived facts are obtained by lifting discretized readings. For instance:

- a kitchen motion trigger at t_1 becomes Motion(*Kitchen*, t_1);
- appliance events may yield Open(*FridgeDoor*, t_2);
- or SwitchOn(*CoffeeMachine*, t_3).

These facts constitute the evidential layer of the inquiry: they describe what was observed, not yet why it occurred.

Higher-level activities are usually inferred. The activity

Cooking(*Alice*, t) (9)

is rarely measured by a dedicated sensor; it can be deduced from co-occurring evidence such as

- InRoom(*Alice*, *Kitchen*, t),

- Using(*Alice*, *Stove*, *t*),
- Using(*Alice*, *Fridge*, *t*),

through a rule of the form

$$\forall p, t \text{ Using}(p, \text{Stove}, t) \wedge \text{Using}(p, \text{Fridge}, t) \rightarrow \text{Cooking}(p, t). \quad (10)$$

Temporal structure can likewise be made explicit. A cooking-related sequence

$$\text{Open}(\text{Fridge}) \rightarrow \text{Take}(\text{Food}) \rightarrow \text{Use}(\text{Stove}) \rightarrow \text{Eat}(\text{Food}) \quad (11)$$

may be accompanied by relations such as

- Before(*t*₁, *t*₂),
- Occurs(Open(*Fridge*), *t*₁).

Contextual conditions such as weekday, time of day, occupancy, or outdoor temperature enter the same language, enabling rules such as

$$\text{Weekday}(d) \wedge \text{Morning}(t) \wedge \text{CoffeeMachineOn}(t) \rightarrow \text{Breakfast}(t). \quad (12)$$

Importantly, the inquiry itself is represented symbolically. For the question “Does opening windows reduce my electricity consumption?”, one may introduce

- Experiment(*E*₁),
- Question(*E*₁, *ReduceConsumptionByWindowOpening*),
- Participant(*E*₁, *Alice*),
- Starts(*E*₁, *t*₁), Ends(*E*₁, *t*₂).

Observations such as

- Open(*Window*, *t*),
- PowerConsumption(*very_high*)

then become evidence for or against a hypothesis, e.g.

$$\text{WindowOpen}(t) \rightarrow \text{LowerHeatingPower}(t). \quad (13)$$

An occupant annotation such as “Alice is preparing dinner” is stored as

$$\text{PreparingDinner}(\text{Alice}, t), \quad (14)$$

not as a disposable training label, but as a reusable assertion in the knowledge base.

This representation changes what annotation means in the system. Labels are no longer opaque class identifiers attached to feature vectors; they are logical statements that can be queried, composed, and checked. For example, $\exists t \text{ Cooking}(\text{Alice}, t)$ asks whether cooking occurred, while

$$\exists t \text{ Cooking}(\text{Alice}, t) \wedge \text{HighPowerConsumption}(t) \quad (15)$$

asks whether cooking coincided with high electricity use—a question directly aligned with energy-oriented inquiry.

A case is not created by FOL; it is detected upstream by *D* (Section 3.1.3) and only represented in FOL once facts about it are formalized. All facts concerning the same case share its identifier, so that they can be retrieved and reasoned about together. For a dishwasher case *c*₄, this may include

- EnergyWh(*c*₄, 450),

- $\text{DurationMinutes}(c_4, 60)$,
- $\text{UsesProgramme}(c_4, \text{Half Load})$,
- $\text{InitialSoil}(c_4, \text{Slightly Dirty})$,
- $\text{WashingResult}(c_4, \text{Clean})$.

The first two facts are lifted directly from sensor-derived indicators; the remaining three become available only once the corresponding annotations have been accepted. The knowledge base as a whole contains many such case-scoped facts together with relations, rules, questions, and experiments; it is not equivalent to any single case.

4.2.3. LLMs and first-order logic

FOL and LLMs play complementary roles. Symbolic reasoners are reliable for deduction and consistency checking, but brittle when evidence is incomplete or when the mapping from everyday language to formal structure is ambiguous. LLMs are strong at interpreting natural language and proposing plausible explanations, but weak at guaranteeing logical soundness. The architecture therefore uses FOL as a shared interface between the two.

Because predicates are structured text, an LLM can read facts such as

- $\text{InRoom}(\text{Alice}, \text{Kitchen}, 08:12)$,
- $\text{SwitchOn}(\text{Kettle}, 08:13)$,
- $\text{Open}(\text{Fridge}, 08:14)$

much more effectively than raw records of the form "sensor": 4, "value": 1. In practice, the reasoning pipeline first converts discretized sensor evidence into logical facts; the LLM then receives those facts together with an occupant question, for example "What activity is most likely occurring?". It may answer abductively—suggesting breakfast from a kitchen–fridge–kettle pattern—without accessing the underlying continuous measurements.

The exchange is bidirectional. From a set of observed facts, the LLM may propose candidate assertions such as

- $\text{Cooking}(\text{Alice}, 08:03)$,
- $\text{PreparingMeal}(\text{Alice}, 08:04)$,

optionally with a confidence score. These candidates are proposals, not accepted knowledge: they enter the knowledge base as accepted facts only once they have been validated symbolically and/or confirmed by the occupant. In this way, language models enrich the inquiry with semantic hypotheses and explanations, while FOL preserves an explicit, inspectable record of what has actually been accepted as knowledge.

4.2.4. Annotation-aiding system architecture

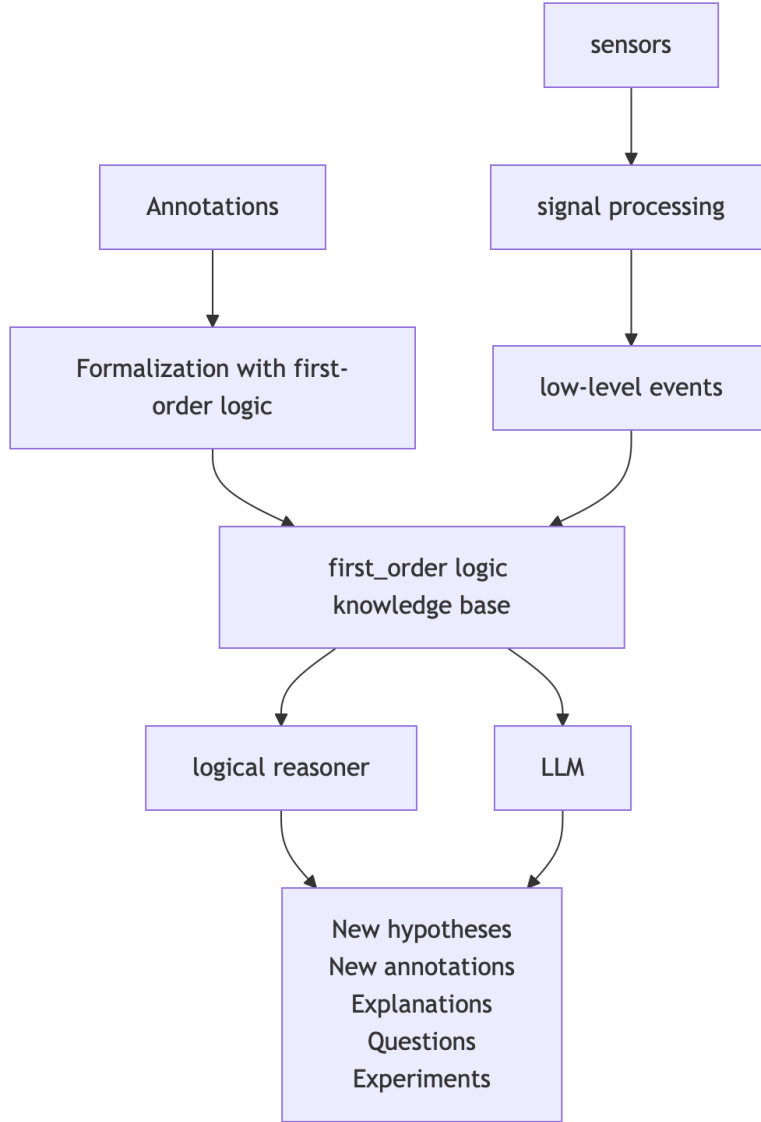


Figure 4: Shared-representation architecture between first-order logic and an LLM. The figure shows how the two components communicate through the logical layer; it does not depict the chronological order of the end-to-end inquiry flow (Figure 2).

Figure 4 summarizes the resulting pipeline. Natural-language utterances from occupants are mapped to logical predicates or goals; sensor streams are discretized and lifted into the same FOL vocabulary; a symbolic reasoner performs deduction and consistency checks; and an LLM supports dialogue, abduction, and explanation. Both components communicate through the shared logical layer rather than through ad hoc text or opaque embeddings alone.

Within this architecture, annotation is a knowledge-acquisition act embedded in the inquiry. Occupants ask questions in natural language; the system gathers the corresponding sensor evidence; missing semantic dimensions are requested when needed; accepted annotations become logical assertions; and the LLM can later explain relations or suggest further experiments. FOL thus operates as a semantic protocol among sensors, reasoners, occupants, and language models, supporting explainable and inquiry-driven smart-home intelligence beyond the conventional use of annotations as ground-truth labels for supervised learning.

4.3. Label Assignment

The second module, *Label Assignment*, acquires or proposes semantic annotations for timeslot-level sensor data through an interactive learning process: for each detected sensing context, an ask/no-ask trigger decides whether to reuse or suggest a supported annotation directly, or to ask the occupant instead—either branch yielding a candidate annotation. Instead of treating annotation as a one-step classification task, this module first determines whether a timeslot's context is already sufficiently supported to be resolved automatically or should generate an interaction with the occupant. If the available information is sufficient, the system proposes the annotation directly and stores the result in the annotation corpus. Otherwise, the timeslot is selected as an *ask*, meaning that additional occupant feedback is required before the candidate annotation can be finalized.

4.3.1. Trigger Mechanism — Ask or not

Within the current scope of this study, we adopt the trigger mechanism proposed by Silva et al. (2022) as the basis for deciding when user feedback should be requested. As illustrated in Fig. 5, the trigger mechanism evaluates whether the current sensing context is informative enough to justify an interaction. The decision can be based on neighborhood-based criteria, such as local density and classifier estimation error, or global criteria, such as spread rate. Label redundancy can be used in both branches to ensure that each label has sufficient support in the knowledge database. These criteria help the system avoid unnecessary questions while still collecting labels for uncertain, sparse, or underrepresented contexts. Table 8 summarizes the trigger criteria, their hyper-parameters, and their corresponding decision branches.

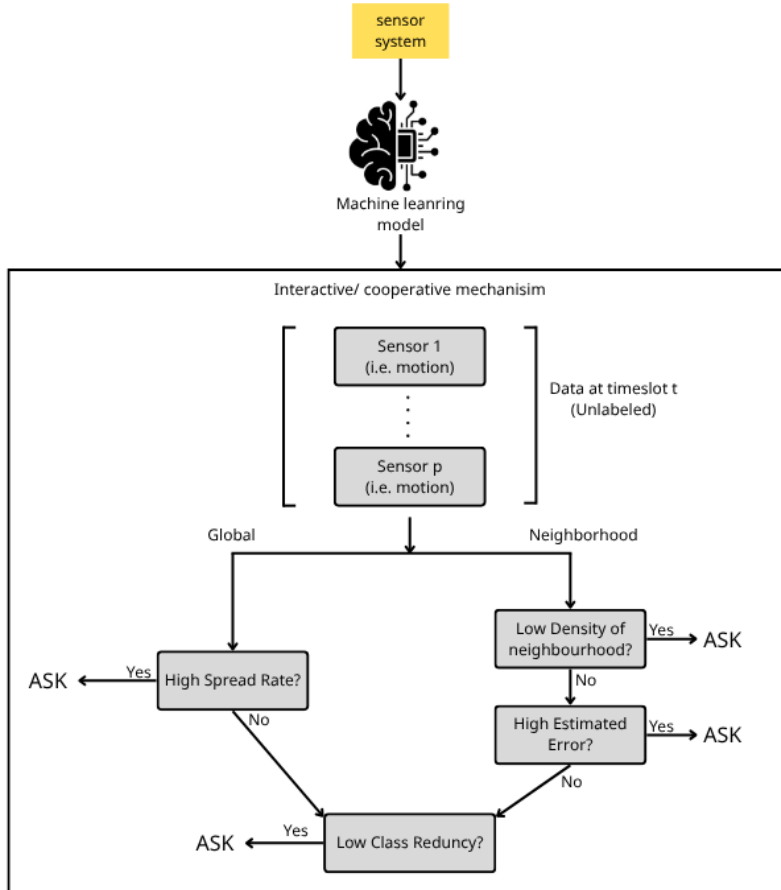


Figure 5: Trigger mechanism for deciding whether a timeslot should generate an ask.

Table 8: Trigger criteria used in the interactive learning mechanism

Criterion	Hyper-parameters	Branch	Meaning
Density of the neighborhood	ϵ ; minimum number of neighbors	Neighborhood	Ask when the current context has too few labelled neighbors. Related to local coverage-based active learning and density-based methods such as DBSCAN.
Classifier estimation error	Maximum acceptable estimation error	Neighborhood	Filters unreliable neighbors based on classifier error. Ensures local consistency and reduces noise.
Label redundancy	Minimum number of occurrences per label	Neighborhood / Global	Ensures that each label has enough examples.
Spread rate	Improvement ratio	Global	Ask when the current sample improves global coverage.

The density-based trigger criterion can be understood through a simple analogy with DBSCAN. As shown in Fig. 6, a point that has enough neighbors can be considered well supported, while a point with too few neighbors may be treated as noise. Similarly, in our annotation system, a sensing context that has not been sufficiently observed or labelled before can be regarded as an unfamiliar point. Instead of discarding it, the system treats it as a candidate that needs confirmation from the occupant. Therefore, when the current timeslot has too few labeled neighbors, an *ask* is generated to verify its semantic meaning and improve the labeled knowledge database.

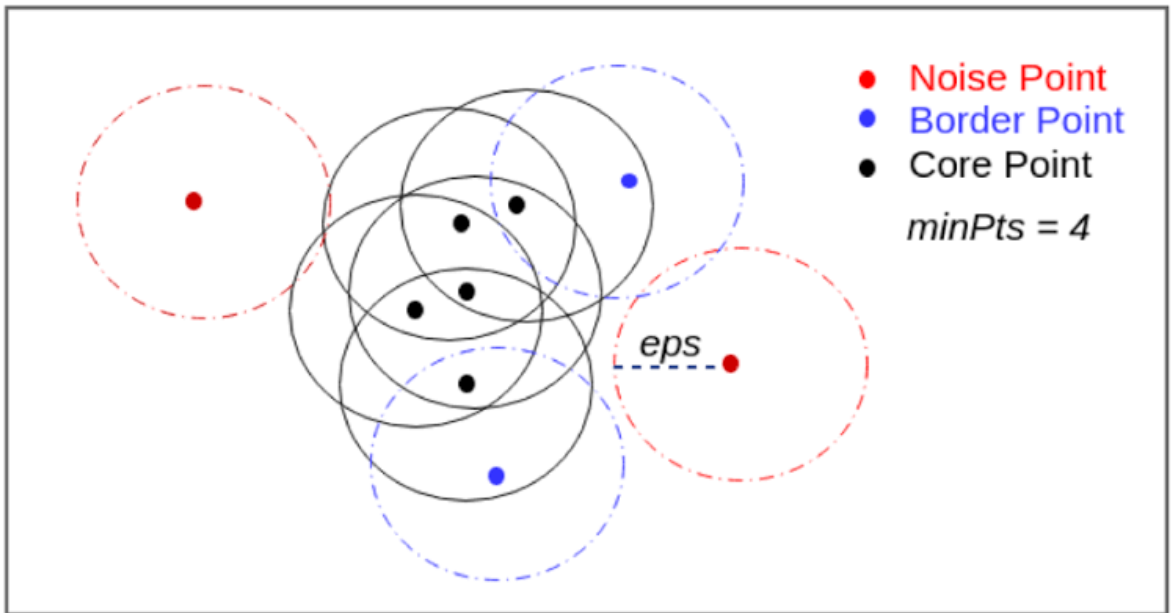


Figure 6: Illustration of the neighborhood density concept in DBSCAN. A low-density point can be interpreted as an unfamiliar context that requires user confirmation.

Among these criteria, the current deployment activates only the density-of-the-neighborhood criterion. Because the annotation target is not a single-label activity class but a structured, partly open-ended set of 5W1H-style semantic fields together with occupant satisfaction, the classifier-based criteria remain less reliable until a larger and more consistent labelled corpus is available. The concrete hyper-parameters of the active criterion are specified together with the experiment in Section 5.

4.3.2. Chatbot Interaction and Annotation Workflow

At the beginning of each day, the module builds an annotation schedule for every active experiment from its asking window, timestep length, and maximum number of daily asks, merging the timeslots of all experiments into a single

daily schedule. When a scheduled timeslot is reached, its sensor readings are aggregated into a timeslot-level feature vector, following the discretization described in Section 4.2.1, and evaluated against the trigger criterion. If the timeslot is selected as an *ask*, the system constructs an annotation request from three sources of information—the target timeslot, the metadata of the sensors involved in the experiment, and the current label state of that timeslot—and the LLM generates a user-facing question sent through the Telegram bot, as shown in Fig. 7.

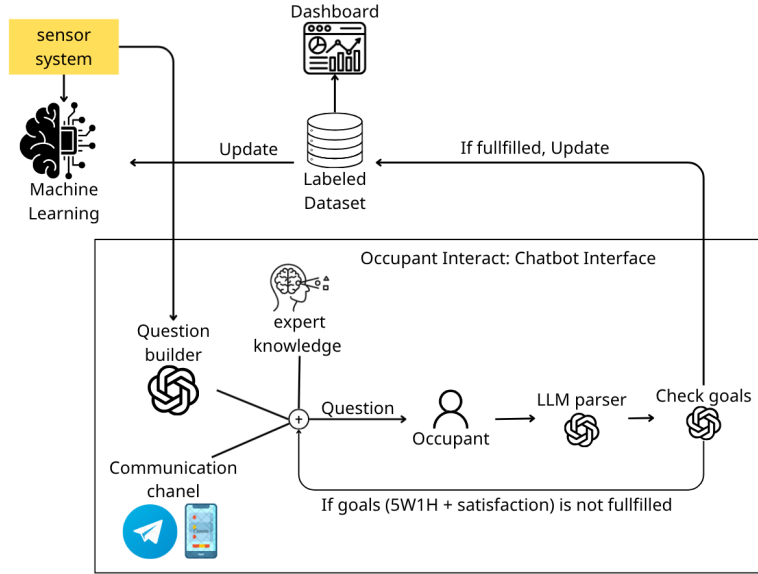


Figure 7: Illustration of user interaction through the Telegram-based annotation interface

As illustrated in Fig. 8, the conversational agent then repeatedly reasons over the current annotation goal, acts by generating a question or parsing the occupant's reply, and observes the updated annotation state, updating the annotation field by field until the required semantic fields are complete. If a timeslot cannot be resolved by the end of the day, it is retained as an unfinished annotation rather than discarded. Completed annotations feed back into the annotation corpus used by the trigger mechanism, so that timeslots resembling previously annotated ones are increasingly resolved automatically and the frequency of occupant interruptions decreases as that corpus grows.

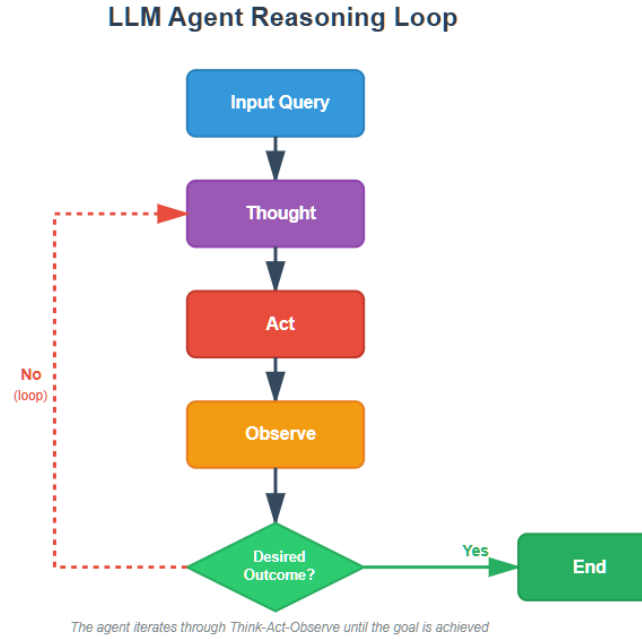


Figure 8: Goal-based reasoning loop followed by the conversational agent during interactive label assignment.

4.3.3. Implementation Setup

This module is implemented as a dedicated service, decoupled from the main backend, so that the interactive learning loop and the resulting annotations remain isolated from the rest of the system. Rather than exchanging data through dedicated API endpoints, the module accesses experiment metadata and sensor readings directly from the shared database and persists annotations to its own store, keeping the label layer separated from the backend. Table 9 summarizes the implementation setup.

Table 9: Implementation setup of the label assignment module.

Component	Configuration
Service framework	FastAPI and Uvicorn.
Runtime environment	Docker with the <code>python:3.12.3-slim</code> image.
ML library	scikit-learn, used for feature scaling and encoding of the labelled dataset.
LLM orchestration	LangChain / LangGraph, driving a goal-based conversational agent.
Language model	<code>gpt-4o-mini</code> , accessed through OpenRouter.
User-facing interface	Telegram Bot.

Each experiment defines the metadata used to schedule and bound the annotation process. The asking window and timestep define when and how often timeslots are evaluated, while the maximum number of asks per day caps how many questions an occupant may receive for a given experiment within a single day. The metadata fields are:

- **Asking window** — the daily time interval during which the system is allowed to send annotation questions to the occupant;
- **Timestep length** — the temporal resolution of the interactive learning process, determining the length of each timeslot and the daily annotation schedule;
- **Maximum asks per day** — the maximum number of questions that may be sent for the experiment within a single day;

- **Selected sensors** — the set of sensors associated with the experiment, defining which sensor readings are retrieved for each timeslot.

Each resolved timeslot is stored as an annotation record that realizes the 5W1H schema of Eq. 5, together with the occupant's satisfaction, a short activity label summarizing the annotated activity, and a completion status. The activity label acts as the target class fed back into the labelled dataset that drives the trigger mechanism. An annotation is only marked complete once its required semantic fields have been filled in; incomplete annotations are retained with this status left unset rather than being discarded.

4.4. Quality Control - Occupant Interaction

Following the principle of inclusive management, a system that supports occupants operates as a cooperative process between humans and machines, in which the aim is not to replace human judgment but to support it (Ploix et al., 2026). Within this cooperation, the system is responsible for collecting and organizing data and for detecting situations that deserve attention, while the occupant is responsible for observing, interpreting, and providing annotations that reflect their own perception of the situation. We organize Occupant Interaction into two complementary channels: Real-time Interaction (Section 4.4.1), conducted through a chat channel, which includes both the system proactively requesting or confirming information and the occupant freely asking questions and receiving explanations from the system; and Reflective Interaction (Section 4.4.2), conducted through a web dashboard, where the occupant independently reviews charts, analyses, and past annotation logs.

Cooperative learning is the strategy that governs this repair: whenever an annotation reaching Quality Control (Section 3.2.3) is found incomplete, uncertain, or conflicting with prior cases, it is not silently accepted or discarded. Instead, the occupant is presented with the annotation and the reason it was flagged, and can confirm it as correct, amend it, reject it, or leave it unresolved for later reconsideration. An unresolved or rejected item can, in turn, generate a new interactive-learning request (Section 4.3) rather than entering the case knowledge silently. This keeps the occupant the authority on household meaning while the system's case knowledge becomes progressively more reliable; it is a repair strategy embedded in Quality Control, not a separate, independently deployed module.

4.4.1. Real-time Interaction

The system sends a real-time interaction whenever a timeslot is left unresolved by Label Assignment (Section 3.2.2) or flagged as invalid by Quality Control (Section 3.2.3), following the interactive and cooperative learning strategies described in del Castillo Cardoso (2025). Each request is anchored to the corresponding sensor context, such as the time, duration, and measurement values involved, and the occupant may also ask why the annotation is needed before responding. When the sensor configuration is not detailed enough to distinguish between appliances sharing the same sensor, for example a TV and a sound system connected through the same smart plug, the system reports this limitation to the occupant, who may choose to report each use individually so the devices can be separated, or accept grouping them under a single label. To keep the level of interaction acceptable over time, requests are only raised when confidence falls below a threshold, are not repeated for previously confirmed situations, and are capped within a given period.

The occupant may also initiate the interaction by freely asking questions through a natural-language interface, about recorded consumption, practical recommendations, or the reasoning behind a given recommendation; in the latter case, the system supports its answer with evidence, such as the number of similar past cases and the average improvement observed among them. Both flows are summarized in Fig. 9.

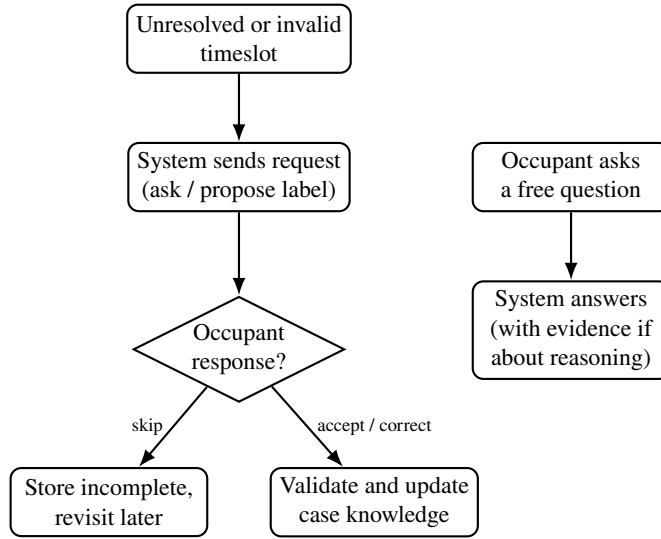


Figure 9: Overview of Real-time Interaction. The system-initiated flow (left) is triggered by Label Assignment or Quality Control; the occupant-initiated flow (right) runs independently through the same chat channel.

4.4.2. Reflective Interaction

The occupant creates an experiment for the devices they wish to monitor and can then access a dashboard to track its progress, including energy consumption statistics, annotation statistics, and a list of pending questions that can be answered directly on the dashboard.

To help the occupant understand their own behavior and how the system supports it, they can review the annotation history, which includes both information inferred by the system and information provided by the occupant, and adjust it when needed. The system also provides a mechanism for comparing usage cases, helping the occupant determine whether a change in behavior achieves the desired outcome.

5. Experiment

This section selects the target activity for evaluating AnnoBot, instantiates the corresponding inquiry and experiment plan, and reports the results obtained so far.

5.1. Activity Selection and Rationale

The choice of the target activity determines which sensor data are collected, which semantic dimensions require annotation, and which questions drive the interaction with occupants. Because AnnoBot collects a structured 5W1H description together with satisfaction rather than a single activity label, the primary selection criterion is the *diversity* of the semantic information an activity gives rise to. We therefore select the activity on two groups of criteria. *Quality* criteria assess how well an activity exercises the system: the diversity of the semantic dimensions it produces, the separation between the sensor-observable fact layer and the part that requires human input, and the support it offers for learning an estimation model. *Feasibility* criteria assess whether the activity can actually be monitored: the availability of sensor infrastructure and of participants are mandatory, while activity frequency and annotation burden are favourable.

Two candidate activities are available in the instrumented building. The *meeting room* targets team meetings and presentations: motion indicates whether the room is occupied, CO₂ reflects how crowded it is, door contact marks entries and exits, sound level distinguishes discussion from silence, and TV power indicates whether the presentation device is on. Most of the semantic information, however—how many people attend, what equipment is used, why the meeting is held, how it is conducted, and how satisfied participants are with the air quality—must be provided by the occupants, which is exactly what creates the diversity to be annotated. The *kitchen* targets cooking or reheating food with a microwave and taking items from a refrigerator: its fact layer reduces to microwave on/off and refrigerator cycles, so information such as the food type, the meal, who is cooking, and the satisfaction must be provided by the occupants.

Table 10 compares the two candidates on the decisive criteria.

Table 10: Comparison of the two candidate activities on the decisive selection criteria.

Criterion	Meeting room	Kitchen
Semantic diversity	Five dimensions (5W1H + satisfaction), all varying strongly across meetings	Five dimensions but narrow variation (who is 1–2 people, why is 1–2 meals per day)
Fact vs. semantic layer	Rich fact layer: occupancy, crowding, entries/exits, discussion, presentation	Thin fact layer: microwave on/off, refrigerator cycles
Estimation model	Number of people from CO ₂ (Jiang et al., 2016); meeting state from motion, sound, and power (Silva et al., 2022)	Only on/off and NILM (Zoha et al., 2012); who/why/satisfaction have no sensor correlate

The meeting room is superior or tied on every criterion. The decisive difference is the separation between the annotation-free and annotation-based layers and the possibility of learning an estimation model: in the meeting room the fact layer is rich enough for the system to recognise an event before asking the occupant, and the semantic dimensions all correlate with the sensors and can thus be learned from the (sensor $\rightarrow$ annotation) pairs; in the kitchen, the who/why/satisfaction dimensions have no sensor correlate to learn from. The meeting room is therefore selected as the deployment target.

For this activity, the annotation target instantiates the timeslot label of Eq. 5: *who* is the number of attendees, *what* the equipment used, *where* the (fixed) meeting room, *when* the duration, *why* the purpose, *how* the conduct, and *satisfaction* the participants' comfort with air quality and temperature. The sensor-derived dimensions (who via CO₂ and motion, what via TV power, when via door and motion) complement the annotation-based ones (why, how, satisfaction).

5.2. Inquiry Questions and Experiment Plan

Following the inquiry tuple of Eq. 2, the meeting-room experiment $E_{\text{meeting}} = \langle \mathcal{I}_{\text{meeting}}, T_{\text{meeting}} \rangle$ groups four inquiries $\mathcal{I}_{\text{meeting}} = \{i_1, i_2, i_3, i_4\}$ under a single experiment-level listening period T_{meeting} spanning two working weeks; two inquiries are answerable from sensor-derived facts alone and two require annotation. They instantiate the abstract examples of Table 2 with concrete meeting-room content, as summarised in Tables 11 and 12.

Table 11: Inquiry questions for the meeting-room experiment.

	Question Q	Goal Γ	Recognition
i_1	“How many people attended, what was the temperature, and which equipment was used?”	Measure occupancy, temperature, equipment	annotation-free
i_2	“Where does the meeting consume the most energy?”	Decompose energy by source	annotation-free
i_3	“With the same number of people and weather, is setpoint A more comfortable and economical than setpoint B?”	Evaluate comfort–energy trade-off	annotation-based
i_4	“Why did yesterday’s meeting consume more energy than usual?”	Explain an energy anomaly	annotation-based

All four inquiries share the same sensor configuration

$$S = \{\text{CO}_2, \text{motion, sound, door, TV power, temperature, humidity}\}$$

and the same detection rule D : a meeting episode is bounded by door and motion transitions, yielding $[t_{\text{start}}, t_{\text{end}}]$ for each occurrence within T_{meeting} . They differ in the indicators I they read from that episode and in the annotation scope A they require, summarised in Table 12.

Table 12: Per-inquiry configuration for the meeting-room experiment (S and D are shared, as described above).

	Indicators I	Annotation scope A
ι_1	Attendee count (from CO ₂ and motion), average temperature, equipment usage	none required; <i>who</i> is additionally annotated to provide ground truth for the attendee-count estimation model
ι_2	Energy consumption decomposed by source over the episode	none
ι_3	Average temperature, humidity, and energy consumption per set-point	<i>what</i> (setpoint used) and <i>satisfaction</i> (perceived comfort)
ι_4	Energy consumption of the episode relative to comparable past episodes	<i>why</i> (occupant's explanation of the anomaly) and <i>satisfaction</i>

The deployment reuses the trigger configuration of Section 4.3, namely the density-of-the-neighborhood criterion with $\epsilon = 0.3$ and $N_{\min} = 3$, with an asking window from 09:00 to 21:00, a 30-minute timestep, and a maximum of 10 asks per day. Across the annotation-based inquiries (ι_3, ι_4), *what*, *why*, and *satisfaction* are mandatory annotated fields, while *who* and *when* are instead inferred from the sensors (the attendee count from CO₂ and motion, the duration from door and motion).

5.3. Data Characteristics

Although the proposed system includes a data collection module for acquiring sensor events from real residential environments, the current implementation could not yet be deployed in an actual household setting due to time constraints. Therefore, in this experiment, we used an existing dishwasher-related dataset to simulate the sensor data that would normally be collected by the data collection module. This allows the experiment to evaluate the downstream workflow of the system, including sensor-event visualization, event selection, and interactive label assignment, under a realistic time-series setting while postponing full real-world deployment to future work.

Following the experiment formulation in Section 3.1, the dishwasher activity use case can be defined as an experiment containing a single inquiry,

$$E_{\text{dishwasher}} = \langle \{t_{\text{dishwasher}}\}, T_{\text{dataset}} \rangle, \quad (16)$$

$$\iota_{\text{dishwasher}} = \langle Q_{\text{dishwasher}}, S_{\text{dishwasher}}, I_{\text{dishwasher}}, D_{\text{dishwasher}}, A_{\text{dishwasher}}, \Gamma_{\text{dishwasher}} \rangle.$$

In this use case, the inquiry question $Q_{\text{dishwasher}}$ is concerned with understanding dishwasher-related activity and its relationship with appliance power consumption. A representative question is:

Which mode of the dishwasher is more cost-effective, and when should I use that mode?

The selected sensor configuration $S_{\text{dishwasher}}$ consists of dishwasher-related sensor signals available in the dataset, mainly the appliance-level power consumption signal represented by `CUI_power_dishwasher` and `dishwasher`. These signals simulate the data that would be collected from a smart plug or appliance-level power meter attached to the dishwasher. The detection rule $D_{\text{dishwasher}}$ identifies a washing cycle from the power signal and returns its $[t_{\text{start}}, t_{\text{end}}]$; the resulting energy consumption is the indicator $I_{\text{dishwasher}}$ (Eq. 3). The experiment-level listening period T_{dataset} corresponds to the observation period covered by the dataset, from 2019-09-01 06:00 to 2019-12-30 22:30, with a 30-minute sampling interval.

The selected dataset represents a kitchen activity scenario related to dishwasher usage. It contains time-series observations sampled every 30 minutes from 2019-09-01 06:00 to 2019-12-30 22:30. The dataset consists of

4,114 records and 11 columns, with no missing values. Each day contains 34 fixed timestamps, corresponding to the interval from 06:00 to 22:30. Thus, the dataset does not represent a complete 24-hour daily sequence, but rather a fixed daytime observation window.

Table 13 summarizes the main characteristics of the dataset used in this experiment.

Table 13: Summary of the dishwasher activity dataset.

Characteristic	Value
Dataset file	CESTAS_kitchen_wash_dishes.csv
Number of records	4,114
Number of columns	11
Observation period	2019-09-01 to 2019-12-30
Number of observed days	121
Sampling interval	30 minutes
Daily observation window	06:00–22:30
Number of time steps per day	34
Missing values	0
Main sensor signal	Dishwasher power consumption
Main activity labels	wash dishes, wash dishes start

5.4. Results

As a preliminary validation of the implemented pipeline, the following results were obtained using a dishwasher dataset before the meeting-room deployment was available.

5.4.1. System Result Visualization

After loading the dishwasher dataset into the implemented system, the simulated sensor events were visualized through the AnnoBot dashboard. Figure 10 shows the result of this process. The dashboard provides an overview of the experiment, including weekly consumption, annotation progress, pending questions, and a daily consumption timeline. In the daily timeline, dishwasher-related signals are displayed as temporal event traces, allowing the user to inspect when the dishwasher was active and how the activity was distributed across the day.

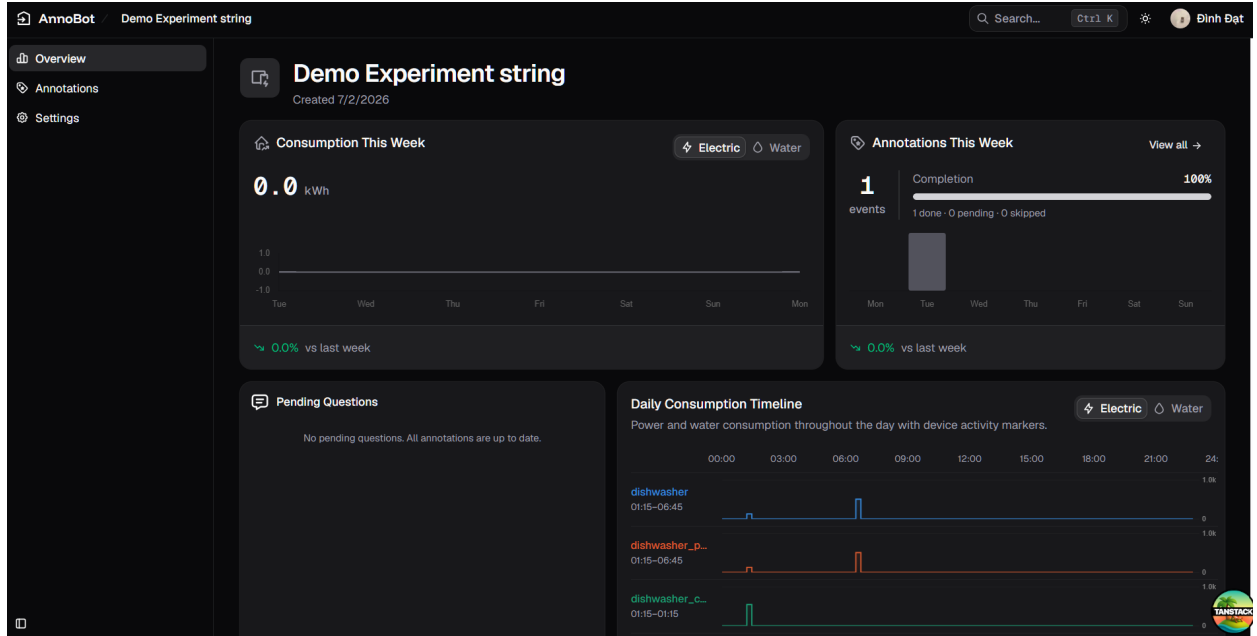


Figure 10: Dashboard result for the dishwasher activity use case after importing the simulated sensor events into AnnoBot.

This result demonstrates that the implemented system can ingest dishwasher-related time-series data, represent it as sensor-event timelines, and present the information in a form that supports annotation and inspection. The dashboard also shows the integration between the simulated data collection output and the annotation management interface. In this example, the system reports one annotation event for the selected week and indicates that the annotation process has been completed. This provides preliminary evidence that the proposed workflow can connect sensor-event visualization with the label assignment process.

5.4.2. Interactive Learning Result

The implemented label assignment module was tested through the Telegram-based chatbot interface. As shown in Fig. 11, the system was able to send annotation questions to the user, receive textual replies, and continue the interaction with follow-up questions. This confirms that the communication loop between the backend, the LLM-based annotation agent, and the Telegram interface operated correctly during testing.

The interaction was stable in both the Telegram mobile application and Telegram Web. Users in the testing phase were able to receive system-generated questions and respond directly through the chat interface. The chatbot also maintained the iterative annotation flow by asking one semantic field at a time, such as the user's activity, reason, manner of performing the activity, and satisfaction state. These observations indicate that the current implementation can support real-time interactive label collection through a familiar messaging interface.

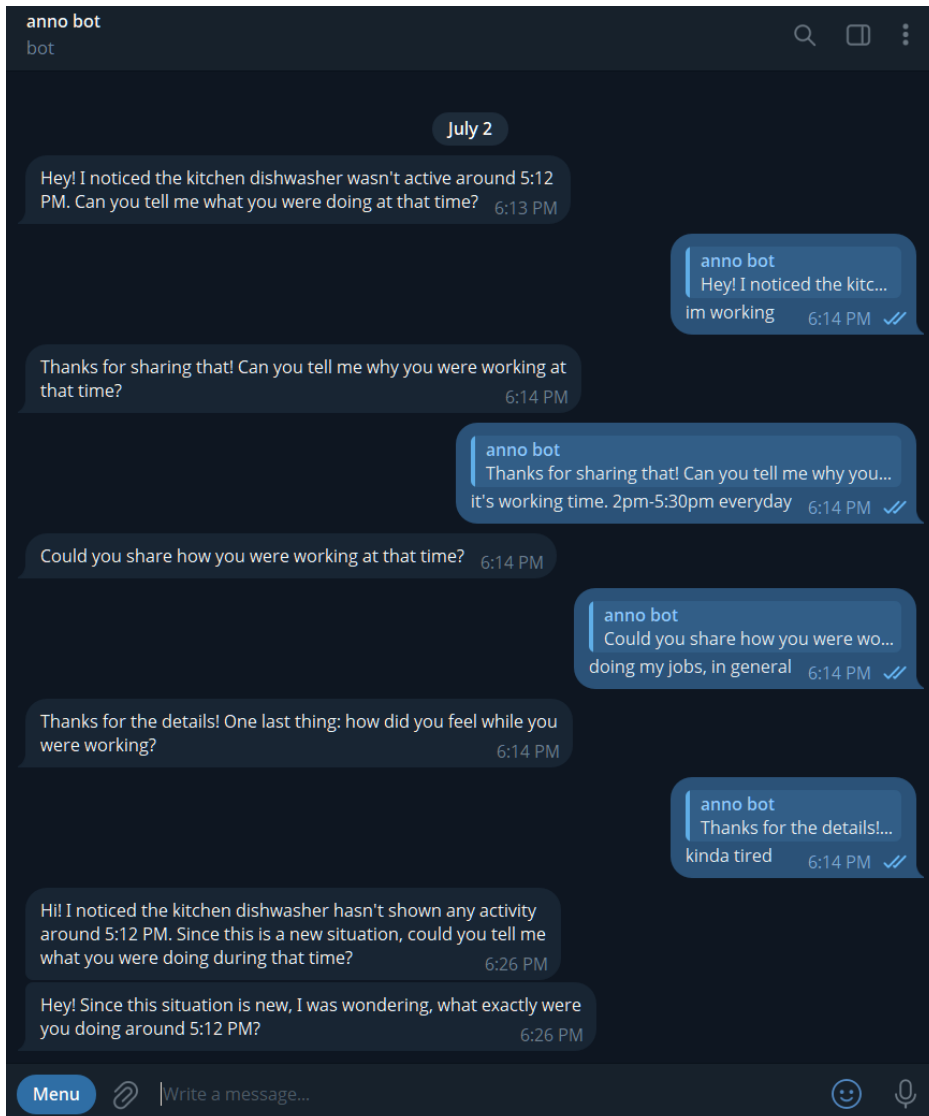


Figure 11: Example of the Telegram-based chatbot interaction during label assignment testing.

5.4.3. Annotation Result

After the system detected or selected a relevant dishwasher-related time slot, the annotation was completed through the chatbot-based label assignment process. The user-provided responses were then stored and displayed in the annotation log interface. Figure 12 shows the resulting annotation record after the interaction was completed.

Time Slot	Activity	What	How	Why	Satisfaction	Status
Invalid Date Invalid Date	working	working	doing my jobs, in general	it's working time	kinda tired	Done

Showing 1 of 1 annotations

< Prev 1 / 1 Next >

Figure 12: Annotation log after completing the chatbot-based label assignment process.

The annotation log presents the semantic labels collected for the selected time slot. Instead of storing only low-level sensor values, the system records user-provided contextual information according to the selected annotation dimensions, including the activity, what the user was doing, how the activity was performed, why it was performed, and the user's satisfaction state. In the illustrated example, the annotation status is marked as *Done*, indicating that the annotation flow was successfully completed and the resulting semantic label was saved by the system.

5.5. Evaluation Metrics

The evaluation follows AnnoBot's goals of accuracy and reduced user burden, drawing on the interactive-learning evaluation of the laboratory's prior work (Amayri et al., 2019; Silva et al., 2022). Table 14 summarises the core metrics.

Table 14: Core evaluation metrics.

Metric	Meaning	Source
Timeslot skip rate	fraction of the annotation burden removed by interactive learning (skipped / total), which should grow over time	annotation statistics
User response rate	whether questions are asked at the right time and are understandable (target > 80%)	AI service log
Questions per annotation	whether the AI asks to the point given the five fields (what/why/how/satisfaction/activity)	conversation log
Sensor data coverage	whether every timeslot has sensor data and the sensors actually report	sensor events
Annotation quality	fraction of timeslots fully annotated, with all fields filled and the activity label set	annotations

6. Conclusion

This paper presented AnnoBot, a smart annotation-aiding system that reframes data annotation for human-centric smart buildings as an occupant-driven inquiry process rather than a step in a machine-learning pipeline. An experiment establishes a time-bounded listening period for one or more inquiries; each inquiry configures the sensors, indicators, detection rule, and annotation scope needed to answer an occupant's question. Selected sensor traces are collected within that listening period, and a detection rule identifies and bounds a relevant service episode. When the resulting sensor facts answer the inquiry question directly, they already constitute a valid, sensor-based case. Otherwise, the system requests the specific semantic dimensions it is missing: interactive learning decides whether the current context is already sufficiently supported or should trigger a question to the occupant, and, within the scope evaluated in this paper, only the local-neighbourhood-density criterion is active for that decision.

Accepted sensor facts and occupant-provided annotations are formalized as first-order-logic assertions sharing a common case identifier, giving the system an explicit, inspectable representation rather than opaque feature vectors. Within this representation, the LLM and the logical reasoner play complementary, bounded roles: the LLM interprets natural language, drives the dialogue, and proposes candidate assertions or explanations, while the logical reasoner checks consistency and completeness; a candidate becomes accepted knowledge only after that symbolic check and/or the occupant's confirmation. Cooperative learning closes this loop as a repair strategy embedded in quality control: incomplete, uncertain, or conflicting annotations are returned to the occupant for confirmation, amendment, rejection, or deferred resolution, rather than being silently accepted or discarded. Accepted cases can then be compared through the inquiry's indicators, revealing patterns that the system explains to the occupant, who may revise a routine or test a new hypothesis, restarting the inquiry loop with new evidence.

The two illustrations used throughout this paper delimit what has actually been evaluated. The meeting-room inquiry plan demonstrates how one experiment can host several concrete inquiries with distinct recognition paths, but has not yet been deployed. The dishwasher dataset, used to validate the implemented pipeline end to end, gives preliminary, component-level evidence that the system can ingest sensor-event time series, visualize them, and collect interactive semantic annotations through a chatbot interface; it does not constitute evidence of longitudinal household validation or of complete sufficiency decision support. Future work should deploy the framework in real households across both illustrative activities, activate and evaluate the remaining interactive-learning trigger criteria, and assess whether the resulting case base measurably supports occupants in understanding and revising their own energy practices.

References

Alemdar, H., Ertan, H., Incel, O.D., Ersoy, C., 2013. Aras human activity datasets in multiple homes with multiple residents, in: Proceedings of the 7th International Conference on Pervasive Computing Technologies for Healthcare, pp. 232–235. doi:10.4108/icst.pervasivehealth.

2013.252120.

- Amayri, M., Ploix, S., Bouguila, N., Würtz, F., 2019. Estimating occupancy using interactive learning with a sensor environment: Real-time experiments. *IEEE Access* 7, 53932–53944. doi:10.1109/ACCESS.2019.2911921.
- Awada, A., Reignier, P., Ploix, S., Jacomino, M., 2020. Identification collaborative d'activités dans une zone habitée. Technical Report. Grenoble INP – Université Grenoble Alpes.
- Bouchabou, D., Nguyen, S.M., Lohr, C., LeDuc, B., Kanellos, I., 2021a. A survey of human activity recognition in smart homes based on IoT sensors algorithms: Taxonomies, challenges, and opportunities with deep learning. *Sensors* 21, 6037. URL: <https://doi.org/10.3390/s21186037>, doi:10.3390/s21186037.
- Bouchabou, D., Nguyen, S.M., Lohr, C., LeDuc, B., Kanellos, I., 2021b. Using language model to bootstrap human activity recognition ambient sensors based in smart homes. *Electronics* 10, 2498. URL: <https://doi.org/10.3390/electronics10202498>, doi:10.3390/electronics10202498.
- del Castillo Cardoso, E.A., 2025. Self-experimentation aiding system for energy management in residential buildings. Ph.D. thesis. Université Grenoble Alpes [2020-....].
- Chatterjee, S., Singh, A., Mitra, B., Chakraborty, S., 2021. Accoustate: Auto-annotation of IMU-generated activity signatures under smart infrastructure. arXiv preprint arXiv:2112.06651 URL: <https://arxiv.org/abs/2112.06651>, arXiv:2112.06651.
- Das, D., Nishimura, Y., Vivek, R.P., Takeda, N., Fish, S.T., Plotz, T., Chernova, S., 2021. Explainable activity recognition for smart home systems. arXiv preprint arXiv:2105.09787 URL: <https://arxiv.org/abs/2105.09787>, arXiv:2105.09787.
- Demrozi, F., Turetta, C., Al Machot, F., Pravadelli, G., Kindt, P.H., 2023. A comprehensive review of automated data annotation techniques in human activity recognition. arXiv preprint arXiv:2307.05988 doi:10.48550/arXiv.2307.05988, arXiv:2307.05988.
- Fiori, M., Civitarese, G., Bettini, C., 2024. Using large language models to compare explainable models for smart home human activity recognition, in: Adjunct Proceedings of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing. URL: <https://doi.org/10.1145/3675094.3679000>, doi:10.1145/3675094.3679000.
- Fiori, M., Civitarese, G., Salim, F.D., Bettini, C., 2026. DomusFM: A foundation model for smart-home sensor data. arXiv preprint arXiv:2602.01910 URL: <https://arxiv.org/abs/2602.01910>, arXiv:2602.01910.
- Fiori, M., Mor, D., Civitarese, G., Bettini, C., 2025. GNN-XAR: A graph neural network for explainable activity recognition in smart homes. arXiv preprint arXiv:2502.17999 URL: <https://arxiv.org/abs/2502.17999>, doi:10.1007/978-3-032-10554-7_19, arXiv:2502.17999.
- Home Assistant, n.d. Home assistant integrations documentation. <https://www.home-assistant.io/integrations/>. Accessed: Jul. 06, 2026.
- HumanSignal, Inc., n.d. Label studio documentation. <https://labelstud.io/guide>. Accessed: Jul. 02, 2026.
- Hussain, Z., Sheng, M., Zhang, W.E., 2019. Different approaches for human activity recognition: A survey. arXiv preprint arXiv:1906.05074 URL: <https://arxiv.org/abs/1906.05074>, arXiv:1906.05074.
- Jiang, C., Masood, M.K., Soh, Y.C., Li, H., 2016. Indoor occupancy estimation from carbon dioxide concentration. *Energy and Buildings* 131, 132–141. doi:10.1016/j.enbuild.2016.09.002.
- Karpekov, A., Iyer, A., Dhekane, S.G., Chernova, S., Plotz, T., 2025. DISCOVER: Data-driven identification of sub-activities via clustering and visualization for enhanced activity recognition in smart homes. arXiv preprint arXiv:2503.01733 URL: <https://arxiv.org/abs/2503.01733>, arXiv:2503.01733.
- Matsui, T., Onishi, K., Misaki, S., Fujimoto, M., Suwa, H., Yasumoto, K., 2020. Salon: Simplified sensing system for activity of daily living in ordinary home. *Sensors* 20, 4895. doi:10.3390/s20174895.
- Ploix, S., Simoiu, M.S., Laurillau, Y., Alvarez Del Castillo Cardoso, E., Boulmaiz, F., Alzouhri Alyafi, A.A., Huang, Y., Reignier, P., 2026. Sufficiency engineering. Working manuscript, Grenoble INP – Université Grenoble Alpes, INVOLVED and LearningHome ANR projects.
- Sager, C., Janiesch, C., Zschech, P., 2021. A survey of image labelling for computer vision applications. *Journal of Business Analytics* 4, 91–110. doi:10.1080/2573234X.2021.1908861.
- Shi, Y., Balasubramanian, A., Herring, J., Li, J., Romero, J.M., Gonzalez, R.S., Mishra, V., Rozga, A., Tan, X.Z., Plotz, T., 2026. TRACE: Temporal reasoning over context and evidence for activity recognition in smart homes. arXiv preprint arXiv:2605.02841 URL: <https://arxiv.org/abs/2605.02841>, arXiv:2605.02841.
- Silva, J.B., Amayri, M., Ploix, S., Reignier, P., Silva, C.S., 2022. Cooperative and interactive learning to estimate human behaviours for energy applications. *Energy and Buildings* 258, 111727. doi:10.1016/j.enbuild.2021.111727.
- Tan, Z., Li, D., Wang, S., Beigi, A., Jiang, B., Bhattacharjee, A., Karami, M., Li, J., Cheng, L., Liu, H., 2024. Large language models for data annotation and synthesis: A survey. arXiv preprint arXiv:2402.13446 doi:10.48550/arXiv.2402.13446, arXiv:2402.13446.
- Thukral, M., Dhekane, S.G., Hiremath, S.K., Haresamudram, H., Plotz, T., 2024. Layout agnostic human activity recognition in smart homes through textual descriptions of sensor triggers (TDOST). arXiv preprint arXiv:2405.12368 URL: <https://arxiv.org/abs/2405.12368>, arXiv:2405.12368.
- Zoha, A., Gluhak, A., Imran, M.A., Rajasegarar, S., 2012. Non-intrusive load monitoring approaches for disaggregated energy sensing: A survey. *Sensors* 12, 16838–16866. doi:10.3390/s121216838.

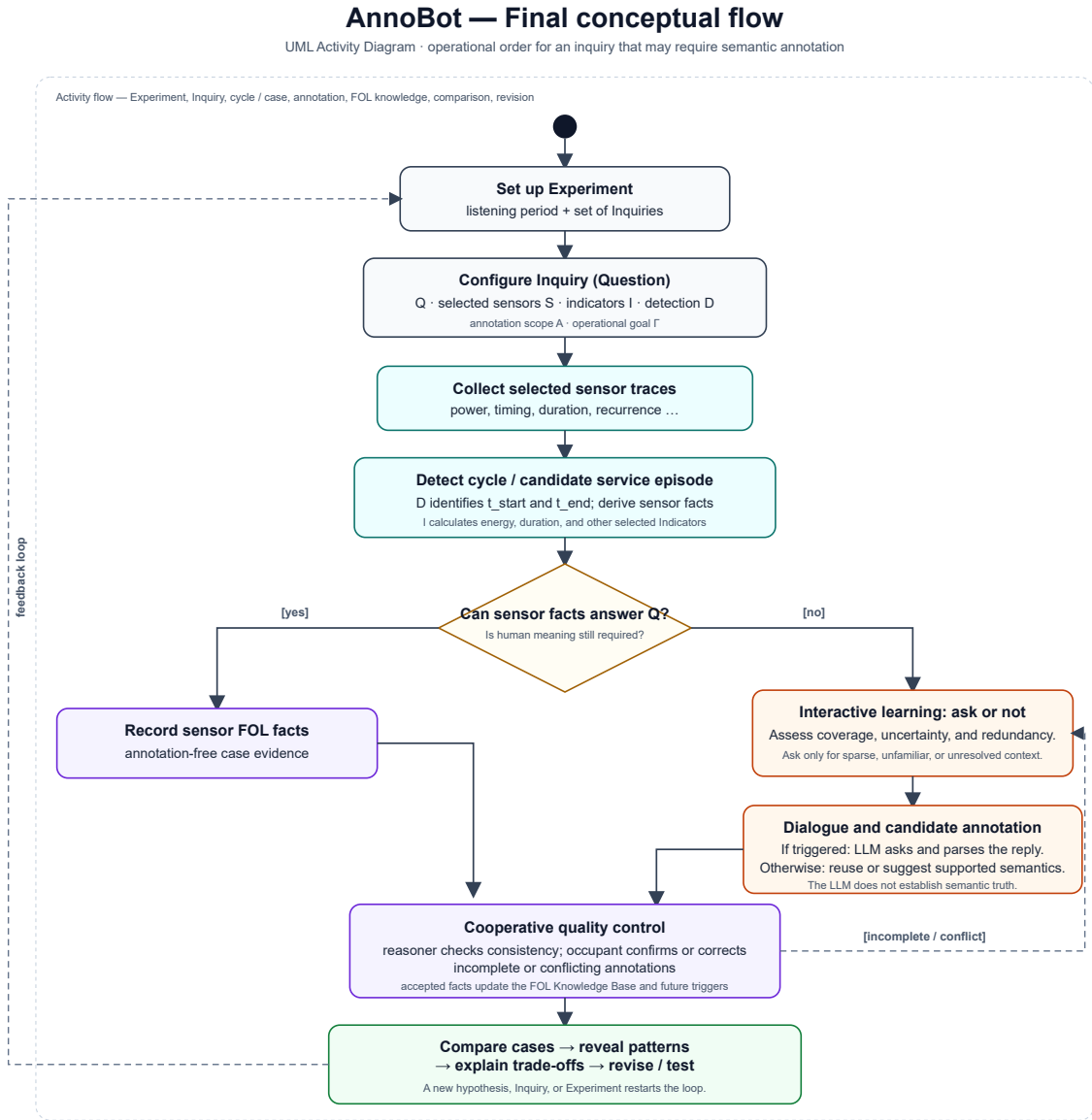


Figure 2: End-to-end inquiry flow. An experiment provides a listening period for configured inquiries; a detected service episode follows either a sensor-only or annotation path. Accepted facts share a first-order-logic representation before cases are compared, explained, and revised.